The Tannakian substrate of GeoVac: a finite-cutoff pro-system, a neutral Tannakian category, and an explicit $\mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ inclusion into $\mathrm{Aut}^{\otimes}(\omega)$

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Abstract

We construct the Tannakian substrate of the GeoVac spectral triple at finite cutoff and verify, bit-exact on explicit `sympy`-rational panels, every prerequisite of the Deligne–Milne 1982 neutral Tannakian theorem on an abelian primitive Hopf algebra $\mathcal{H}_{\mathrm{GV}}(n_{\max}) = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ whose primitive space $V_{n_{\max}}$ has $\mathbb{Q}$-dimension $3N(n_{\max})$ (one generator per (sector, Mellin slot) pair). The construction has two halves.

Pro-system half. A cofiltered system of finite-cutoff truncations $\{\mathcal{H}_{\mathrm{GV}}(n_{\max})\}_{n_{\max}\in\mathbb{N}}$ with explicit Hopf-algebra transition maps $\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(k)$ (block-diagonal truncations) satisfies the cofiltered axiom, is U^*-equivariant, admits a well-defined inverse limit $\mathcal{O}_\infty \cong \mathbb{Q}^{\mathbb{N}_{\mathrm{sec}}}$ into which the continuum cochains χ_∞, η_∞ and the continuum Mellin lift $F(s)$ of v3.66.0 FO1–FO3 project coherently, and has finite-cutoff endomorphism algebra $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}})$ that is block-lower-triangular of closed-form dimension $\dim = \sum_{j\le i\le n_{\max}}(i+1)(j+1)$.

Tannakian half. The finite-dimensional representation category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ is an abelian, symmetric monoidal, rigid $\mathbb{Q}$-linear category (axioms TC-1a, TC-1b, TC-1c bit-exact); the no-op forgetful $\omega :\ \mathrm{Rep}_{\mathrm{fin}} \to \mathbf{Vec}_{\mathbb{Q}}$ is a faithful exact $\otimes$-functor (TC-1d); and the candidate cosmic-Galois group $U^*_{\mathrm{Levi}} = \mathbb{G}_a^{3N(n_{\max})} \rtimes \mathrm{SL}_2$ admits an *explicit inclusion*

$$\Phi :\ U^*_{\mathrm{Levi}}(\mathbb{Q}) \longrightarrow \mathrm{Aut}^{\otimes}(\omega), \qquad \Phi(t,g)(V) = \exp\Big(\sum_{g\in\mathfrak{P}} t_g X_g^V\Big) \otimes \rho_{V_{\mathrm{PW}}}(g)$$

on the natural-substrate panel $V \otimes V_{\mathrm{PW}}$, which is bit-exact at every point of the verification grid (TC-1e + TC-1f). At $n_{\max} = 2$ the inclusion is *full-panel injective*: for every one of the $3 \cdot N(2) = 15$ primitive generators g a 2-dim test rep V_g detects g uniquely with $\eta_{V_g}(t) = I$ iff $t_g = 0$.

Total bit-exact verification panel: 5,864 *zero residuals* (`sympy`-rational throughout) across PS-1/2/3/4 + TC-1a/b/c/d/e/f + TC-2a/b/c/d + G4-Inj at $n_{\max} \in \{2,3,4\}$. The G4-Inj block (2,904 of these) holds by construction (multiplicativity of a ring map, a coproduct identity on the GeoVac side, the condition $\det g = 1$ on its SL_2 panel, and panel-size bookkeeping), so those residuals do not certify the comparison theorem (Theorem 6.2). The fourteenth math.OA standalone in the GeoVac series.

Cosmic-Galois comparison homomorphism. Sending each M3 generator to the depth-1 de Rham class of its level-4 motivic period, and each M1 and M2 generator to 0, gives a Hopf map out of the GeoVac Hopf algebra; dually it is a homomorphism *from* the unipotent radical $\mathcal{U}_4$ of the level-4 motivic Galois group $\mathcal{G}_4 = \mathcal{G}_{\mathrm{MT}(\mathbb{Z}[i,1/2])}$ of Deligne 2010 + Glanois 2015 *into* the abelian factor $\mathbb{G}_a^{3N}$ of GeoVac's group, factoring through $\mathcal{U}_4^{\mathrm{ab}}$, with image of

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dimension at most 2 per weight (Section 6, Theorem 6.2). The proof uses the Eskandari–Murty–Nemoto 2025 level-4 placement of G + the Deligne 2010 faithful action (level 4) on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_4)$. Depth-$\geq 2$ content, which a primitive Hopf substrate cannot receive, is the named long-range forward research question.

Honest scope. We close the *inclusion* direction $\mathrm{Aut}^{\otimes}(\omega) \supseteq U^*_{\mathrm{Levi}}$ at finite cutoff with both Levi factors explicit (TC-1e/TC-1f). The *converse equality* $\mathrm{Aut}^{\otimes}(\omega) = U^*_{\mathrm{Levi}}$ at finite cutoff is closed by the TC-2 sub-arc: TC-2a (abelian factor at $n_{\max} = 2$), TC-2b (SL_2 factor on the Peter–Weyl panel), TC-2c (higher cutoffs $n_{\max} \in \{3, 4\}$), and TC-2d (pro-system cofiltered coherence on the natural-substrate panel). The remaining long-range content is whether an enriched (non-primitive, depth ≥ 2) GeoVac Hopf structure receives more of $\mathcal{U}_4$ than its abelianization — the pro-finite question coherent with v3.66.0 FO3 Interpretation C on $\mathcal{O}_\infty$ (PS-3).

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1 Introduction

1.1 The question

The GeoVac spectral triple is constructed on a finite combinatorial graph at every cutoff $n_{\max} \in \mathbb{N}$: a Coulomb graph on Fock-projected S^3 (Paper 7) carrying a Camporesi–Higuchi spinor bundle with half-integer-shifted Dirac spectrum (Paper 32). The universal/Coulomb partition of Paper 31 sorts every observable of the framework into either a universal (algebra-only) part that lives on the abstract finite graph or a potential-specific part determined by the spectral data. After the Mixed-Tate Test of v3.46.0 + the Periods-A arc of v3.48.0 + the Q5'-HardParts arc of v3.61.0–v3.66.0 the structural reading of the universal sector crystallised: each of the three master-Mellin engine sub-mechanisms M1 / M2 / M3 lives in a cyclotomic mixed-Tate period ring at level ≤ 4 over $\mathbb{Z}[i, 1/2]$; the natural categorical home for these periods is a (motivic) Galois group, and the natural candidate is a pro-unipotent affine group scheme U^* with Levi decomposition $U^*_{\mathrm{Levi}} = \mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ at finite cutoff (v3.63.0 Track 1).

The present paper asks the next structural question: is the substrate for a *neutral Tannakian theorem* on the underlying Hopf-algebra category fully assembled at finite cutoff? Concretely:

- Is there a cofiltered pro-system $\{\mathcal{H}_{\mathrm{GV}}(n_{\max}), \Phi_{m,k}\}$ of Hopf algebras whose inverse limit carries the period content?
- Is the representation category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ abelian, symmetric monoidal, and rigid (the three Deligne–Milne 1982 prerequisites for being neutral Tannakian)?
- Is there a fiber functor $\omega : \mathrm{Rep}_{\mathrm{fin}} \to \mathbf{Vec}_{\mathbb{Q}}$?
- Does the candidate $U^*_{\mathrm{Levi}} = \mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ admit an *explicit inclusion* into $\mathrm{Aut}^{\otimes}(\omega)$?

This paper answers *yes* to all four, bit-exact at finite cutoff, on the abelian primitive substrate $\mathcal{H}_{\mathrm{GV}}(n_{\max}) = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ where $V_{n_{\max}}$ has $\mathbb{Q}$-dimension $3N(n_{\max})$.

1.2 What this paper does not claim

The Tannakian reconstruction direction $\mathrm{Aut}^{\otimes}(\omega) = U^*_{\mathrm{Levi}}$ at finite cutoff is closed in Sections 5.5, 5.6, 5.7, and 5.8 as the TC-2 sub-arc (Theorems 5.7, 5.9, 5.11, 5.13). What is *not* closed by this paper is the *inverse-limit identification* of the unipotent part $\mathbb{G}_a^{\infty}$ of $\mathrm{Aut}^{\otimes}(\omega)^{(\infty)} = \mathbb{G}_a^{\infty} \rtimes SL_2$ with the motivic Galois group on the mixed-Tate period ring of v3.66.0 FO3 Interpretation C on $\mathcal{O}_{\infty}$ (PS-3) — this requires the pro-finite Tannakian theorem [2, 3] coherent with the period content and remains the named long-range frontier.

We also do *not* introduce any new ring of periods, any new spectral object, or any new conjecture about α. The Hopf algebra $\mathcal{H}_{\mathrm{GV}}$ is the v3.61.0 Track A Hopf candidate; the period content is the v3.66.0 FO1–FO3 + Paper 55 census; the cosmic-Galois candidate U^*_{Levi} is the v3.63.0 Track 1 Levi decomposition. This paper consolidates the substrate for these existing pieces into one bit-exact assembly.

1.3 Roadmap

Section 2 constructs the pro-system half (PS-1 transitions + PS-2 U^*-equivariance + PS-3 inverse limit + PS-4 endomorphism rigidity). Section 3 verifies that $\mathrm{Rep}_{\mathrm{fin}}$ is abelian, symmetric monoidal, and rigid (TC-1a/b/c). Section 4 constructs the fiber functor (TC-1d). Section 5 establishes the explicit $\mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ inclusion into $\mathrm{Aut}^{\otimes}(\omega)$ (TC-1e + TC-1f), plus the TC-2 converse-direction sub-arc closing the finite-cutoff reconstruction in Sections 5.5 (abelian factor), 5.6 (SL_2 factor), 5.7 (higher cutoffs), and 5.8 (cofiltered coherence). Section 6 gives the comparison with the level-4 motivic Galois group $\mathcal{G}_4$: a homomorphism from its unipotent radical $\mathcal{U}_4$ into the abelian factor of U^*_{GV}, through $\mathcal{U}_4^{\mathrm{ab}}$ (Theorem 6.2). Section 7 catalogues the 5,864-residual bit-exact verification panel. Section 8 states the open long-range frontier (reconstruction) and the natural sprint-scale follow-ons (higher $n_{\max}$ injectivity panels, SL_2 on higher Peter–Weyl reps, non-natural-substrate enlargements).

1.4 Related work

GeoVac's gauge-structure scaffolding is the Marcolli–van Suijlekom 2014 gauge-network construction [10] as sharpened by Perez-Sanchez 2024/2025 [11, 12] — the Standard Model recovered from the spectral action without a Higgs (matching the GeoVac Sprint H1 Yukawa-non-selection theorem). *Honest scope of the MvS-lineage claim:* GeoVac is an *instance* of MvS 2014 (with the Perez-Sanchez 2025 correction applied) at the data level (graph, vertex spectral triple, edge connection, spectral action). It is an *extension* of MvS at the structural level: the forced-graph claim (Bertrand × Fock projection), the master Mellin engine, the explicit Hopf-algebra substrate $\mathcal{H}_{\mathrm{GV}}$, and the Tannakian dual U^* constructed here are GeoVac-original. MvS 2014 does *not* predict a Tannakian dual; the cosmic-Galois layer U^*_{Levi} sits on top of the gauge-network scaffolding as additional structure, not as an MvS corollary.

The neutral-Tannakian backbone we transport is Deligne–Milne 1982 [1], the modern motivic-Galois treatment is André 2004 [4] and Brown [5], the cyclotomic mixed-Tate package is Deligne 2010 [7] + Glanois 2015 [8], and the Connes–Marcolli cosmic Galois group U^*_{CM} is [9]. We flag one common framing slip: the two groups are related by a *surjection* $U^*_{CM} \twoheadrightarrow \mathcal{G}_{MT(\mathbb{Z})}$, not an identification. This is an elementary comparison of two free pro-unipotent groups, not a theorem we are importing: U^*_{CM} has one generator in every positive degree (free non-abelian, Connes–Marcolli [9]) while $\mathcal{G}_{MT(\mathbb{Z})}$ has generators only in odd degrees ≥ 3 (Brown 2012 [5]). Brown does *not* assert the comparison; his cosmic-Galois paper explicitly leaves open whether his groups are related to the Connes–Marcolli ones at all. The period ring of GeoVac sits inside $\mathrm{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4; see Paper 55 [23] for the cyclotomic-mixed-Tate census. The natural cosmic-Galois group to compare with U^*_{GV} is therefore $\mathcal{G}_{MT(\mathbb{Z}[i,1/2])} = \mathcal{G}_4$, the level-4 motivic Galois group (Deligne 2010), *not* Brown's smaller $\mathcal{G}_{MT(\mathbb{Z})}$ (Eskandari–Murty–Nemoto 2025 [25] place Catalan G at level 4; that G is *not* a period of $\mathrm{MT}(\mathbb{Z})$ is expected but not proven, so the refinement is motivated by where G is known to live, not compelled by a proven exclusion). Bost–Connes / Greenfield–Marcolli–Teh QSM lineage is *ruled out* as a transport target on four structural grounds (see Paper 55 §subsec:open_m2_m3 honest-scope paragraph and the Greenfield–Marcolli–Teh transport memo); the substrate built here is neutral Tannakian on a finite-dimensional symmetric algebra, not QSM.

2 The pro-system half

2.1 The cutoff Hopf algebra $\mathcal{H}_{\mathrm{GV}}(n_{\max})$

For each integer $n_{\max} \geq 1$ let $\mathfrak{S}(n_{\max}) = \{(n, l) : 1 \leq n \leq n_{\max}, 0 \leq l \leq n\}$ be the set of *Coulomb sectors at cutoff* $n_{\max}$ (the construction's convention, matching `geovac/pro_system.py`), $N_{\mathrm{sec}}(n_{\max}) = |\mathfrak{S}(n_{\max})| = n_{\max}(n_{\max}+3)/2$. The *primitive space* of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ is the $\mathbb{Q}$-vector

space

$$V_{n_{\max}} = \bigoplus_{(n,l)\in\mathfrak{S}(n_{\max})} \bigoplus_{k\in\{0,1,2\}} \mathbb{Q}\cdot x_{(n,l),k},$$

of $\mathbb{Q}$-dimension $3N_{\mathrm{sec}}(n_{\max})$, where $k \in \{0,1,2\}$ is the *master Mellin slot*: $k=0$ for the volume-slot mechanism M1, $k=1$ for vertex-parity Hurwitz M3, $k=2$ for Seeley–DeWitt M2 (Paper 18 §III.7; Paper 32 §VIII).

Definition 2.1. The *GeoVac Hopf algebra at cutoff* $n_{\max}$ is $\mathcal{H}_{\mathrm{GV}}(n_{\max}) = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$, the symmetric algebra of the primitive space, with coproduct $\Delta x = x\otimes 1 + 1\otimes x$ on generators (so every generator is primitive), counit $\varepsilon x = 0$, antipode $Sx = -x$, and extended multiplicatively. This is the v3.61.0 Track A candidate Hopf algebra.

$\mathcal{H}_{\mathrm{GV}}(n_{\max})$ is a primitively generated commutative Hopf algebra over $\mathbb{Q}$ (a polynomial Hopf algebra). Its group of group-like elements is trivial; its Lie algebra of primitive elements is the abelian Lie algebra $V_{n_{\max}}$.

2.2 Transition maps and the cofiltered axiom (PS-1)

Definition 2.2. For $m \geq k \geq 1$ let $\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(k)$ be the block-diagonal Hopf-algebra truncation that sends a generator $x^{(m)}_{(n,l),s}$ to $x^{(k)}_{(n,l),s}$ if $n \leq k$ and to 0 otherwise, extended multiplicatively.

Theorem 2.3 (Cofiltered axiom, PS-1)**.** *For every $m \geq k \geq j \geq 1$ the composition $\Phi_{k,j}\circ\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(j)$ equals the direct transition $\Phi_{m,j} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(j)$ bit-exact on every element of $\mathcal{H}_{\mathrm{GV}}(m)$.*

Proof. On a primitive generator $x^{(m)}_{(n,l),s}$, both sides act as truncation by $n \leq j$: $\Phi_{m,j}(x) = x^{(j)}_{(n,l),s}$ if $n \leq j$, else 0; $\Phi_{k,j}(\Phi_{m,k}(x))$ first truncates by $n \leq k$ then by $n \leq j$, which collapses to $n \leq j$ since $j \leq k$. Extension to $\mathcal{H}_{\mathrm{GV}}(m) = \mathrm{Sym}_{\mathbb{Q}}(V_m)$ is via the multiplicative structure, preserved by definition. Bit-exact verification at all 10 triples in $\{1,\ldots,5\}$ confirms the identity at the matrix level. See PS-1 sprint memo. □ □

2.3 U^*-equivariance (PS-2)

The cosmic-Galois candidate U^*_{Levi} acts on $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ at the primitive level via the Levi decomposition $\mathbb{G}_a^{3N(n_{\max})} \rtimes \mathrm{SL}_2$. The pro-unipotent $\mathbb{G}_a^{3N(n_{\max})}$ factor acts as $x \mapsto x + t_x\cdot 1$ on each generator (translation, where $t_x \in \mathbb{Q}$); the SL_2 factor acts on the Mellin slot via a $\mathbb{Q}$-linear action on the $k=0,1,2$ index (the v3.63.0 substrate explicitly identifies the SL_2 factor as acting on the $j_{\max}$ axis of the truncation, with Peter–Weyl decoration on the representations realising the action; see §5.2).

Theorem 2.4 (U^*-equivariance, PS-2)**.** *The transition $\Phi_{m,k}$ is U^*-equivariant: for every $t = (t_x) \in \mathbb{Q}^{3N(m)}$ and every $g \in \mathrm{SL}_2(\mathbb{Q})$, the identity*

$$\Phi_{m,k}\circ\Phi(t,g) \;=\; \Phi(t_{|k},g)\circ\Phi_{m,k} \quad \text{on } \mathcal{H}_{\mathrm{GV}}(m),$$

holds bit-exact. Here $t_{|k}$ is the restriction of t to the primitive generators of $\mathcal{H}_{\mathrm{GV}}(k)$. Verified at 435 $\mathbb{G}_a$ generator compatibility cells +40 class-level χ,η compatibility cells; SL_2 commutativity is categorical (independence of axes).

2.4 The inverse limit $\mathcal{O}_\infty$ (PS-3)

Let $\mathcal{O}_{n_{\max}}$ be the (abelian) class algebra of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ at cutoff $n_{\max}$ (one class per sector). The transition $\Phi_{m,k}$ restricts to a sector-truncation $\mathcal{O}_m \to \mathcal{O}_k$. Set

$$\mathcal{O}_\infty = \varprojlim_{n_{\max}} \mathcal{O}_{n_{\max}}.$$

Theorem 2.5 (Inverse limit, PS-3). *$\mathcal{O}_\infty \cong \mathbb{Q}^{\mathbb{N}_{\mathrm{sec}}}$, the $\mathbb{Q}$-vector space of finitely-supported sequences over the natural-number-indexed sector set $\mathbb{N}_{\mathrm{sec}} = \{(n,l) : n \geq 1, 0 \leq l \leq n\}$ (shell n therefore has dimension $n+1$, giving $N(n_{\max}) = n_{\max}(n_{\max}+3)/2$; this is the convention `geovac/pro_system.py` enumerates, and the one Theorem 2.6's $(i+1)\times(j+1)$ blocks require). The continuum cochains χ_∞, η_∞ of v3.59.0 (Mellin M2-residue $\sqrt{\pi}/2$ + sector-LOCAL closed forms) project coherently to $\mathcal{O}_\infty$: for every $n_{\max}$, the projection $\pi_{n_{\max}} : \mathcal{O}_\infty \to \mathcal{O}_{n_{\max}}$ sends χ_∞ to $\chi_{n_{\max}}$, and similarly for η_∞. The continuum Mellin lift $F(s)$ of v3.65.0 T1 (cf. v3.66.0 FO2) projects with weight/depth grading on $MT(\mathbb{Q},1)$ onto each $\mathcal{O}_{n_{\max}}$.*

The proof is the universal property of the inverse limit applied to the cofiltered system $\{\Phi_{m,k}\}$ of Theorem 2.3, together with the bit-exact projection identities of PS-3 (284 residuals on the universal property and continuity-under-transitions panels).

2.5 Endomorphism rigidity (PS-4)

Theorem 2.6 (Endomorphism rigidity, PS-4). *Let $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}})$ denote the algebra of $\mathbb{Q}$-linear endomorphisms of $\mathcal{O}_{n_{\max}}$ that intertwine all transitions $\Phi_{m,k}$ for $m,k \leq n_{\max}$. Then $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}})$ is the block-lower-triangular subalgebra of $\mathrm{End}_{\mathbb{Q}}(\mathcal{O}_{n_{\max}})$ with respect to the natural sector grading, of closed-form dimension*

$$\dim_{\mathbb{Q}} \mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}}) = \sum_{1\leq j\leq i\leq n_{\max}} (i{+}1)(j{+}1) = \frac{n_{\max}(n_{\max}+1)(n_{\max}+2)(3n_{\max}+13)}{24} \quad (= 19, 55, 125, 2$$

The closed form is a symbolic identity—proved, and regression-pinned, in `tests/test_paper56_endo_rigidity`

Remark 2.7. The Tannakian-naive expectation would be that $\mathrm{End}_{\mathrm{compat}}(\mathcal{O}_{n_{\max}}) = \mathbb{Q}\cdot\mathrm{id}$ (rigid scalar diagonal). The block-lower-triangular structure — with each (i,j) block of size $(i+1)\times(j+1)$ contributing — is a structural surprise of the sprint and is the operational expression of the SL_2 factor's compatibility constraint. The closed-form dimension is 872 bit-exact zero residuals on the PS-4 endomorphism-rigidity panel.

2.6 Pro-system half summary

Sub-track	Statement	Residuals
PS-1	Transitions $\Phi_{m,k}$ + cofiltered axiom	130
PS-2	U^*-equivariance ($\mathbb{G}_a$ + class-level)	485
PS-3	Inverse limit + continuity	284
PS-4	Endomorphism rigidity	872
Total pro-system half		**1,771**

Table 1: Pro-system half: 1,771 bit-exact zero residuals across the four lockdown sub-tracks.

3 The representation category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$

3.1 Finite-dimensional Hopf representations

Definition 3.1. A *finite-dimensional representation* V of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ over $\mathbb{Q}$ is a pair $(\underline{V}, \{X_g^V\})$ where $\underline{V}$ is a finite-dim $\mathbb{Q}$-vector space and $\{X_g^V \in \mathrm{End}_{\mathbb{Q}}(\underline{V})\}_{g\in\mathfrak{P}}$ is a family of *pairwise commuting nilpotent* endomorphisms indexed by the primitive generator set $\mathfrak{P}$ of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$. The action of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ on $\underline{V}$ is by the algebra homomorphism $x_g \mapsto X_g^V$, extended multiplicatively (well-defined because the X_g^V commute).

The category $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ has these representations as objects and *intertwiners* $f : V \to W$ (i.e. $\mathbb{Q}$-linear maps satisfying $f \circ X_g^V = X_g^W \circ f$ for every g) as morphisms.

3.2 Abelian (TC-1a)

Theorem 3.2 (TC-1a)**.** $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ *is an abelian $\mathbb{Q}$-linear category in the Deligne–Milne 1982 sense: it has a zero object, finite direct sums, kernels and cokernels, every monomorphism is the kernel of its cokernel, and every epimorphism is the cokernel of its kernel.*

The proof inherits the abelian structure of $\mathbf{Vec}_{\mathbb{Q}}$ at the level of underlying vector spaces and checks that the kernel / cokernel of an intertwiner inherits a commuting-nilpotent endomorphism family — which it does, because the X_g^V are sub- and quotient-stable under intertwiners. Bit-exact verification on a 5-rep × 5-morph panel at $n_{\max} \in \{2, 3\}$ across the six universal-property axioms (zero / $\oplus$ / ker / coker / mono = ker(coker) / epi = coker(ker)) gives 106/106 zero residuals.

3.3 Symmetric monoidal (TC-1b)

Theorem 3.3 (TC-1b)**.** $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ *is a strict symmetric monoidal category with unit object the trivial 1-dim rep* $\mathbf{1}$ *(all $X_g^{\mathbf{1}} = 0$).*

The tensor structure is diagonal: $V \otimes W$ has underlying space $\underline{V} \otimes_{\mathbb{Q}} \underline{W}$ and endomorphisms $X_g^{V\otimes W} = X_g^V \otimes I_{\underline{W}} + I_{\underline{V}} \otimes X_g^W$ (the cocommutative Hopf-algebra coproduct). Associativity is the identity in lex order; braiding is the swap permutation $V \otimes W \to W \otimes V$, symmetric. Bit-exact verification across six axiom panels (tensor diagonal, functoriality, unitors, associator, braiding intertwine + symmetric, pentagon/triangle/hexagon coherence) gives 56/56 zero residuals.

3.4 Rigid (TC-1c)

Theorem 3.4 (TC-1c)**.** $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$ *is rigid: every V has a dual $V^\vee$ with evaluation* $\mathrm{ev} : V^\vee \otimes V \to \mathbf{1}$ *and coevaluation* $\mathrm{coev} : \mathbf{1} \to V \otimes V^\vee$ *satisfying the snake identities.*

The dual is constructed via the contragredient action: $V^\vee$ has underlying space the $\mathbb{Q}$-dual $\underline{V}^*$, with endomorphisms $X_g^{V^\vee} = -(X_g^V)^T$. The evaluation and coevaluation are the standard pairings on $\mathbf{Vec}_{\mathbb{Q}}$; they intertwine the X_g actions because $X_g^{V^\vee\otimes V} + X_g^{V\otimes V^\vee}$ vanishes on the image of coev and on the kernel of ev.

Remark 3.5 (Strict double dual)**.** On the abelian primitive substrate $(V^\vee)^\vee = V$ is a *strict equality*, not merely a canonical isomorphism: $X_g^{(V^\vee)^\vee} = -(-X_g^{VT})^T = X_g^V$, and the double-dual underlying space identification $\underline{V}^{**} = \underline{V}$ holds strictly on the finite-dim panel. The Hopf antipode squares to identity, $S^2 = \mathrm{id}$, in the symmetric algebra of an abelian Lie algebra, which is the structural reason. **Scope:** this strict equality should not be claimed beyond the abelian primitive substrate $\mathcal{H}_{\mathrm{GV}} = \mathrm{Sym}_{\mathbb{Q}}(V)$.

Bit-exact verification gives 50/50 zero residuals on the rigidity panel.

4 The fiber functor (TC-1d)

Definition 4.1. The *fiber functor* $\omega:\ \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max})) \to \mathbf{Vec}_{\mathbb{Q}}$ sends $V = (\underline{V}, \{X_g^V\})$ to $\underline{V}$ (the underlying $\mathbb{Q}$-vector space) and an intertwiner f to f itself.

Theorem 4.2 (TC-1d)**.** *ω is a faithful exact $\mathbb{Q}$-linear $\otimes$-functor $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max})) \to \mathbf{Vec}_{\mathbb{Q}}$:*

1. *(exact) $\omega(\ker f) = \ker \omega(f)$ and $\omega(\mathrm{coker}\, f) = \mathrm{coker}\, \omega(f)$ bit-exact (the kernel/cokernel calculation in $\mathrm{Rep}_{\mathrm{fin}}$ is the kernel/cokernel in $\mathbf{Vec}_{\mathbb{Q}}$ on underlying spaces);*
2. *(faithful) $f \neq 0$ in $\mathrm{Rep}_{\mathrm{fin}}$ iff $\omega(f) \neq 0$ in $\mathbf{Vec}_{\mathbb{Q}}$ (intertwiners are determined by their underlying linear maps);*
3. *($\otimes$-preserving) $\omega(V \otimes W) = \omega(V) \otimes_{\mathbb{Q}} \omega(W)$ strict;*
4. *(unit-preserving) $\omega(\mathbf{1}) = \mathbb{Q}$.*

Bit-exact verification across the four properties + a faithfulness witness panel gives 40/40 zero residuals.

Remark 4.3. The fiber functor here is the *no-op forgetful*: a finite-dimensional Hopf representation IS its underlying vector space plus the Hopf-data decoration $\{X_g^V\}$, and ω forgets the decoration. This is the natural choice for an abelian primitive Hopf algebra over $\mathbb{Q}$; in the Deligne–Milne 1982 vocabulary ω is a $\mathbb{Q}$-rational fiber functor and the resulting Tannakian category is *neutral*.

5 The explicit $\mathbb{G}_a^{3N} \rtimes \mathrm{SL}_2$ inclusion

5.1 The pro-unipotent factor (TC-1e)

Theorem 5.1 (TC-1e)**.** *The map*

$$\Phi:\ \mathbb{Q}^{3N(n_{\max})} \to \mathrm{Aut}^{\otimes}(\omega), \qquad \Phi(t)(V) = \exp\Big(\sum_{g \in \mathfrak{P}} t_g X_g^V\Big) = \eta_V(t),$$

is a well-defined natural $\otimes$-automorphism of ω for every $t \in \mathbb{Q}^{3N(n_{\max})}$, and a group homomorphism into $\mathrm{Aut}^{\otimes}(\omega)$.

Proof sketch. The matrix exponential terminates exactly because $\sum_g t_g X_g^V$ is a $\mathbb{Q}$-linear combination of pairwise commuting nilpotent endomorphisms, hence nilpotent. The four Deligne–Milne 1982 natural $\otimes$-automorphism axioms reduce to:

- *Invertibility.* $\eta_V(t)^{-1} = \eta_V(-t)$, since exp is a group homomorphism from $\mathbb{Q}^{3N(n_{\max})}$ to $\mathrm{GL}(\underline{V})$ on a commuting nilpotent input.
- *Unit.* $\eta_{\mathbf{1}}(t) = \exp(0) = I$ on the trivial rep (all $X_g^{\mathbf{1}} = 0$).
- *Naturality.* For an intertwiner $f:\ V \to W$, $f \circ X_g^V = X_g^W \circ f$ extends to $f \circ \eta_V(t) = \eta_W(t) \circ f$ by termwise application of the exp power series.
- *$\otimes$-compatibility.* $\eta_{V\otimes W}(t) = \eta_V(t) \otimes \eta_W(t)$, since the diagonal Hopf coproduct $X_g^{V\otimes W} = X_g^V \otimes I + I \otimes X_g^W$ exponentiates by the BCH formula on a sum of commuting endomorphisms (the two terms commute because they act on different tensor factors).

The group-law condition $\Phi(t_1 + t_2) = \Phi(t_1) \cdot \Phi(t_2)$ is the commuting-nilpotent BCH again. Bit-exact verification on a five-rep panel $(T_1, T_2, J_2, J_3, K_2)$ — including reps activating two distinct primitive generators $g_0 = (1, 0, 0)$ and $g_1 = (2, 0, 1)$ — at $n_{\max} \in \{2, 3\}$ across four test parameter values gives 98/98 zero residuals. □ □

5.2 The SL_2 factor (TC-1f)

Theorem 5.2 (TC-1f, SL_2 axioms on Peter–Weyl panel). *On the Peter–Weyl panel*

$$\mathfrak{PW} = \{V_{\mathrm{triv}},\ V_{\mathrm{fund}} = \mathbb{Q}^2,\ \mathrm{Sym}^2 V_{\mathrm{fund}} = \mathbb{Q}^3\}$$

with the standard $\mathrm{SL}_2(\mathbb{Q})$ *action* $\rho:\ \mathrm{SL}_2(\mathbb{Q}) \to \mathrm{GL}(V_{\mathrm{PW}})$ *(identity on* V_{triv}*;* $g \mapsto g$ *on* V_{fund}*;* $g \mapsto \mathrm{Sym}^2(g)$ *on* $\mathrm{Sym}^2 V_{\mathrm{fund}}$*), the three structural identities* ρ *is required to satisfy — invertibility,* $\otimes$*-compatibility, group homomorphism — are bit-exact for every* g *in the five-element panel* $\{e, u, t, s, w\}$ = {*identity, unipotent* $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$*, torus* $\mathrm{diag}(2, 1/2)$*, generic* $\left(\begin{smallmatrix}5 & 2\\ 7 & 3\end{smallmatrix}\right)$*, Weyl involution* $\left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)$}*.*

Theorem 5.3 (TC-1f, $\mathbb{G}_a \times \mathrm{SL}_2$ commute on combined reps). *For every* $t \in \mathbb{Q}^{3N(n_{\max})}$*, every* $g \in \mathrm{SL}_2(\mathbb{Q})$*, every* $V \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max}))$*, and every* $V_{\mathrm{PW}} \in \mathfrak{PW}$*, the two automorphisms of* $\omega(V \otimes V_{\mathrm{PW}}) = \underline{V} \otimes_{\mathbb{Q}} \underline{V_{\mathrm{PW}}}$ *commute:*

$$(\eta_V(t) \otimes I)(I \otimes \rho_{V_{\mathrm{PW}}}(g)) \ = \ \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g) \ = \ (I \otimes \rho_{V_{\mathrm{PW}}}(g))(\eta_V(t) \otimes I).$$

Proof. Both sides are the Kronecker product $\eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g)$ by the Kronecker identity $(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$, which is the structural witness that $\mathbb{G}_a^{3N}$ and SL_2 act on disjoint tensor factors of the combined rep. The combined automorphism is $\Phi(t, g)(V \otimes V_{\mathrm{PW}}) = \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g)$. □ □

Bit-exact verification on 100 (parameter, g, V, V_{PW}) cells gives 100/100 zero residuals.

5.3 Full-panel injectivity at $n_{\max} = 2$ (TC-1f)

Theorem 5.4 (TC-1f, full-panel injectivity). *At* $n_{\max} = 2$*, for every one of the* $3 \cdot N(2) = 15$ *primitive generators* $g \in \mathfrak{P}_{n_{\max}=2}$ *there is a 2-dim test representation* $V_g \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ *with*

$$X_g^{V_g} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad X_h^{V_g} = 0 \text{ for every } h \neq g,$$

such that

$$\eta_{V_g}(t) = I \quad \Longleftrightarrow \quad t_g = 0.$$

Consequently the inclusion $\Phi:\ \mathbb{Q}^{3N(2)} \to \mathrm{Aut}^{\otimes}(\omega)$ *is injective at* $n_{\max} = 2$*: if* $t \neq 0$ *then* t *has some* $t_{g_*} \neq 0$ *and* $\eta_{V_{g_*}}(t) \neq I$*, so* $\Phi(t) \neq \mathrm{id}_\omega$*.*

Proof. $\eta_{V_g}(t) = \exp(t_g X_g^{V_g}) = I + t_g X_g^{V_g}$ since $(X_g^{V_g})^2 = 0$. The $(1, 2)$ entry of $\eta_{V_g}(t) - I$ equals t_g, which is zero iff $t_g = 0$. Bit-exact verification on the full 255-cell panel (15 zero-t cells + 15 single-t activations × 15 test reps + 15 generic-t cells) gives 255/255 zero residuals. □ □

5.4 Inclusion summary

Corollary 5.5 (TC-1e + TC-1f). *At every* $n_{\max} \geq 1$ *there is an explicit group homomorphism*

$$\Phi:\ U^*_{\mathrm{Levi}}(\mathbb{Q}) = \mathbb{G}_a^{3N(n_{\max})}(\mathbb{Q}) \rtimes \mathrm{SL}_2(\mathbb{Q}) \longrightarrow \mathrm{Aut}^{\otimes}(\omega),$$

$$\Phi(t, g)(V \otimes V_{\mathrm{PW}}) = \eta_V(t) \otimes \rho_{V_{\mathrm{PW}}}(g),$$

on the natural-substrate panel $\{V \otimes V_{\mathrm{PW}}:\ V \in \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(n_{\max})),\ V_{\mathrm{PW}} \in \mathfrak{PW}\}$*. At* $n_{\max} = 2$ *the* $\mathbb{G}_a$ *factor of* Φ *is injective on the full primitive panel.*

Remark 5.6 (Scope)**.** Corollary 5.5 closes the *inclusion direction* $\mathrm{Aut}^{\otimes}(\omega) \supseteq U^*_{\mathrm{Levi}}$ at finite cutoff with both Levi factors explicit. The full converse equality $\mathrm{Aut}^{\otimes}(\omega) = U^*_{\mathrm{Levi}}$ on *both* Levi factors is the Deligne–Milne 1982 Theorem 2.11 Tannakian reconstruction direction and would require a separate argument that no other natural $\otimes$-automorphisms of ω exist on the combined $n_{\max}$-axis $\times$ $j_{\max}$-axis substrate. The *abelian-factor* converse at finite cutoff is, however, closeable bit-exactly: this is the content of Section 5.5 below (Theorem 5.7). Full-panel injectivity at $n_{\max} \geq 3$ is sprint-scale (the construction is identical, only the panel size grows by $3N(n_{\max})$); extension of SL_2 to Peter–Weyl reps $\mathrm{Sym}^k V_{\mathrm{fund}}$ for $k \geq 3$ is likewise sprint-scale.

5.5 Finite-cutoff equality on the abelian factor (TC-2a)

We can close the converse equality on the *abelian factor* $\mathbb{G}_a^{3N(n_{\max})} \subseteq \mathrm{Aut}^{\otimes}(\omega)$ at finite cutoff bit-exactly. The construction is direct: parameterise every candidate natural $\otimes$-automorphism on the faithful witness panel by symbolic $\mathbb{Q}$-indeterminates and read off the dimension of the resulting bit-exact linear variety. This is the content of Sprint Q5'-Tannakian-Closure TC-2a, 2026-06-06; memo `debug/sprint_q5p_tc2a_aut_equality_memo.md`, data `debug/data/sprint_q5p_tc2a_aut_equali`
driver `debug/compute_q5p_tc2a_aut_equality.py`, tests `tests/test_tannakian_aut_equality.py`.

Theorem 5.7 (TC-2a, finite-cutoff equality on the abelian factor)**.** *At $n_{\max} = 2$, restrict the fiber functor ω to the $n_{\max}$-axis substrate $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ (forgetting the Peter–Weyl decoration). The bit-exact $\mathbb{Q}$-linear variety of natural $\otimes$-automorphisms of ω on the faithful witness panel of Theorem 5.4 together with the unit object $\mathbf{1}$ has dimension*

$$\dim \mathrm{Aut}^{\otimes}(\omega)\big|_{n_{\max}=2,\ \text{abelian factor}} \;=\; 15 \;=\; 3N(2) \;=\; \dim \mathbb{G}_a^{3N(2)}(\mathbb{Q}).$$

Equivalently, the closed-form solution of the naturality system on the witness panel is

$$\eta_{V_g} = \begin{pmatrix} 1 & q_g \\ 0 & 1 \end{pmatrix} = \exp(q_g\, E_{12}) \qquad (\, q_g \in \mathbb{Q} \text{ free, one per primitive generator}\,), \qquad \eta_{\mathbf{1}} = 1,$$

matching $\Phi(t)(V_g)$ from Section 5.1 with $t_g = q_g$ bit-exactly on all 15 panel reps. The TC-1e/TC-1f inclusion $\mathbb{G}_a^{3N(2)} \subseteq \mathrm{Aut}^{\otimes}(\omega)$ is therefore an equality on the abelian factor at this cutoff.

Proof. Parameterise η_{V_g} on each of the 15 witness reps of Theorem 5.4 by four symbolic $\mathbb{Q}$-indeterminates (p_g, q_g, r_g, s_g), plus one $\eta_{\mathbf{1}}$ indeterminate for the unit object. Total: 61 indeterminates. For every ordered pair (V_a, V_b) in the panel extended by $\mathbf{1}$, compute the $\mathbb{Q}$-basis of $\mathrm{Hom}(V_a, V_b)$ by solving the intertwining linear system $f\, X_g^{V_a} = X_g^{V_b}\, f$ (for every primitive generator g) bit-exactly. The panel has 271 morphisms: 30 endomorphisms ($\dim \mathrm{End}(V_g) = 2$ for each g: centraliser of E_{12} in $M_2(\mathbb{Q})$), 210 cross-pair morphisms ($\dim \mathrm{Hom}(V_g, V_h) = 1$ for $g \neq h$: matrices of the form $\begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}$), 15 morphisms $\mathbf{1} \to V_g$ (dim = 1: matrices $(a, 0)^T$), 15 morphisms $V_g \to \mathbf{1}$ (dim = 1: matrices $(0, a)$), and the identity $\mathbf{1} \to \mathbf{1}$. Impose naturality $\eta_{V_b} f = f\, \eta_{V_a}$ for every basis morphism; at the matrix-entry level this produces 735 $\mathbb{Q}$-linear equations (dropping trivially-zero entries). Append the unit normalisation $\eta_{\mathbf{1}} = 1$ as the 736th equation. The resulting linear system $A\, v = b$ over $\mathbb{Q}$ has shape $A \in \mathbb{Q}^{736 \times 61}$, rank 46, and is consistent ($\mathrm{rank}([A \mid b]) = \mathrm{rank}(A) = 46$). The nullity is $61 - 46 = 15$. Solving in closed form: each panel rep solves to $p_g = 1, r_g = 0, s_g = 1$ with q_g free, and $\eta_{\mathbf{1}} = 1$. The 15 free parameters $\{q_g\}_g$ are in bijection with the 15 primitive generators of $\mathcal{H}_{\mathrm{GV}}(2)$. The closed-form $\eta_{V_g} = \exp(q_g\, E_{12}) = \Phi((\ldots, t_g = q_g, \ldots))(V_g)$ matches the TC-1e inclusion of Section 5.1 on all 15 reps bit-exact. □ □

Remark 5.8 (Scope of TC-2a)**.** Theorem 5.7 closes the converse direction on the *abelian factor* $\mathbb{G}_a^{3N(2)}$ at the single cutoff $n_{\max} = 2$. The SL_2 factor of U^*_{Levi} acts on the orthogonal Peter–Weyl

decoration ($j_{\max}$ axis, Section 5.2) and is not detected by the $n_{\max}$-axis-only fiber functor used here; it requires the combined panel $V \otimes V_{\mathrm{PW}}$. The natural sprint-scale follow-on (TC-2b) layers in the Peter–Weyl panel of Theorem 5.2, parameterises $\eta_{V_g \otimes V_{\mathrm{PW}}}$ on combined reps, imposes naturality against the SL_2-intertwiners from Theorem 5.2 as well as the $\mathcal{H}_{\mathrm{GV}}$-intertwiners, and verifies $\dim = 18 = \dim U^*_{\mathrm{Levi}}$. Higher $n_{\max}$ extends both panels and re-verifies; the PS-2 cofiltered functoriality of Section 2.3 packages the per-cutoff verdicts into a pro-system coherence statement that, together with period-level coherence on $\mathcal{O}_\infty$ (PS-3) per v3.66.0 FO3 Interpretation C, would close the full long-range reconstruction. TC-2a is the first quantitative measurement of the converse direction; it is the abelian-factor sub-claim at $n_{\max} = 2$, not the full claim.

5.6 Finite-cutoff equality on the SL_2 factor (TC-2b)

The SL_2 factor of U^*_{Levi} is reconstructed at the Peter–Weyl panel level via a structurally distinct mechanism: naturality alone gives zero constraint on irreducible reps (Schur's lemma: $\mathrm{End}(V) = \mathbb{Q} \cdot I$ on each irreducible), so the binding constraint comes from *tensor compatibility* on $V_{\mathrm{fund}} \otimes V_{\mathrm{fund}}$ combined with its decomposition isomorphism $V_{\mathrm{fund}} \otimes V_{\mathrm{fund}} \cong \mathrm{Sym}^2 V_{\mathrm{fund}} \oplus V_{\mathrm{triv}}$. Sprint Q5'-Tannakian-Closure TC-2b, 2026-06-06; memo `debug/sprint_q5p_tc2b_sl2_aut_equality_memo.md`, data `debug/data/sprint_q5p_tc2b_sl2_aut_equality.json`, driver `debug/compute_q5p_tc2b_sl2_aut_equ` tests `tests/test_tannakian_sl2_aut_equality.py`.

Theorem 5.9 (TC-2b, finite-cutoff equality on the SL_2 factor)**.** *On the Peter–Weyl panel* $\{V_{\mathrm{triv}}, V_{\mathrm{fund}}, \mathrm{Sym}^2 V_{\mathrm{fund}}\}$ *with the explicit* $\mathbb{Q}$*-linear decomposition isomorphism* $\Phi : V_{\mathrm{fund}} \otimes V_{\mathrm{fund}} \xrightarrow{\sim} \mathrm{Sym}^2 V_{\mathrm{fund}} \oplus V_{\mathrm{triv}}$, $\dim \mathrm{Aut}^{\otimes}(\omega)\big|_{\mathfrak{PW}} = 3 = \dim SL_2$. *Tensor compatibility produces the bit-exact identity*

$$\Phi \cdot (\eta_{V_{\mathrm{fund}}} \otimes \eta_{V_{\mathrm{fund}}}) \cdot \Phi^{-1} = \mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}}) \oplus \det(\eta_{V_{\mathrm{fund}}}),$$

which forces $\eta_{\mathrm{Sym}^2} = \mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}})$ *(top-left* 3×3 *block) and* $\eta_{V_{\mathrm{triv}}} = \det(\eta_{V_{\mathrm{fund}}})$ *(bottom-right* 1×1 *block). Unit normalisation* $\eta_{V_{\mathrm{triv}}} = 1$ *then forces* $\det(\eta_{V_{\mathrm{fund}}}) = 1$*, i.e.* $\eta_{V_{\mathrm{fund}}} \in SL_2(\mathbb{Q})$.

Proof. Parameterise $\eta_{V_{\mathrm{fund}}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ symbolically over $\mathbb{Q}$. The standard Sym^2 formula (`geovac.tannakian._sl` used in TC-1f Theorem 5.2) gives $\mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}})$ as the 3×3 matrix

$$\mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}}) = \begin{pmatrix} a^2 & 2ab & b^2 \\ ac & ad + bc & bd \\ c^2 & 2cd & d^2 \end{pmatrix}.$$

The decomposition isomorphism in basis $\{e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2\}$ (source) and $\{e_1^2, e_1 e_2, e_2^2, e_1 \wedge e_2\}$ (target) is the explicit 4×4 rational matrix

$$\Phi = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1/2 & -1/2 & 0 \end{pmatrix}, \qquad \Phi^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

`sympy`-symbolic computation gives

$$\Phi \cdot (\eta_{V_{\mathrm{fund}}} \otimes \eta_{V_{\mathrm{fund}}}) \cdot \Phi^{-1} = \begin{pmatrix} a^2 & 2ab & b^2 & 0 \\ ac & ad + bc & bd & 0 \\ c^2 & 2cd & d^2 & 0 \\ 0 & 0 & 0 & ad - bc \end{pmatrix},$$

in which the top-left 3×3 block is precisely $\mathrm{Sym}^2(\eta_{V_{\mathrm{fund}}})$ and the bottom-right 1×1 block is $\det(\eta_{V_{\mathrm{fund}}}) = ad - bc$; off-diagonal blocks vanish identically. With unit normalisation $\eta_{V_{\mathrm{triv}}} = 1$,

the bottom-right block forces $ad - bc - 1 = 0$, a single polynomial constraint on the 4-dim ambient $M_2(\mathbb{Q})$. The Jacobian $\nabla(ad - bc - 1) = (d, -c, -b, a)$ has rank 1 at every SL_2-point (verified at $\eta = I$: $(1, 0, 0, 1)$, rank 1; at generic $\eta = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$: $(1, -1, -1, 2)$, rank 1). The variety has codim 1, $\dim = 4 - 1 = 3 = \dim SL_2$. □ □

Corollary 5.10 (TC-2a + TC-2b)**.** *At $n_{\max} = 2$ on the combined natural substrate (the $n_{\max}$-axis $\mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ tensored with the Peter–Weyl panel),* $\dim \mathrm{Aut}^{\otimes}(\omega)\big|_{\mathrm{combined}} = 15+3 = 18 = \dim U^*_{\mathrm{Levi}}$. *The TC-1e/TC-1f inclusion $U^*_{\mathrm{Levi}} \subseteq \mathrm{Aut}^{\otimes}(\omega)$ is therefore an equality at finite cutoff.*

Proof. By the exterior-tensor-product theorem for neutral Tannakian categories [2], for two neutral Tannakian categories C_1, C_2 over $\mathbb{Q}$ with fiber functors ω_1, ω_2, $\mathrm{Aut}^{\otimes}(\omega_1 \boxtimes \omega_2) = \mathrm{Aut}^{\otimes}(\omega_1) \times \mathrm{Aut}^{\otimes}(\omega_2)$. Applying this to $C_1 = \mathrm{Rep}_{\mathrm{fin}}(\mathcal{H}_{\mathrm{GV}}(2))$ with ω_1 the TC-1d no-op forgetful, and C_2 the PW sub-category of $\mathrm{Rep}(SL_2)$ generated by $\{V_{\mathrm{triv}}, V_{\mathrm{fund}}, \mathrm{Sym}^2\}$ with ω_2 the analogous forgetful, the per-factor dims of Theorems 5.7 and 5.9 give $\dim(15) + \dim(3) = 18 = \dim U^*_{\mathrm{Levi}}$. □

Provenance caveat. This corollary's only external warrant is the exterior-tensor-product theorem for neutral Tannakian categories. We cite Deligne's *Catégories tannakiennes* for it, but have not been able to verify a specific numbered statement against the primary text; the section pointer previously given here is withdrawn as unverified. The factorization $\mathrm{Aut}^{\otimes}(\omega_1 \boxtimes \omega_2) = \mathrm{Aut}^{\otimes}(\omega_1) \times \mathrm{Aut}^{\otimes}(\omega_2)$ is standard, but a reader wanting the citable form should consult the primary source directly. □

5.7 Higher cutoffs and the $n_{\max}$-uniform pattern (TC-2c)

The TC-2a abelian-factor argument is parametric in $n_{\max}$: the witness panel grows by $3N(n_{\max})$ generators at each cutoff, the constraint count grows polynomially, and the rank-nullity pattern is $n_{\max}$-uniform. Sprint Q5'-Tannakian-Closure TC-2c, 2026-06-06; memo `debug/sprint_q5p_tc2c_higher_cuto` driver `debug/compute_q5p_tc2c_higher_cutoff.py`, tests `tests/test_tannakian_higher_cutoff.py`.

Theorem 5.11 (TC-2c, finite-cutoff equality at higher cutoffs)**.** *For every $n_{\max} \in \{1, 2, 3, 4\}$,* $\dim \mathrm{Aut}^{\otimes}(\omega)\big|_{n_{\max}\text{-axis}} = 3N(n_{\max})$ *bit-exactly, with the closed-form solution $\eta_{V_g} = \exp(q_g\, E_{12})$ on every panel rep matching $\Phi(t)(V_g)$ from Section 5.1 via the bijection $t_g = q_g$.*

Proof (parametric extension of Theorem 5.7). The argument of Theorem 5.7 is uniform in $n_{\max}$: the witness panel $\{V_g\}_{g \in P^{(n_{\max})}}$ scales with $|P^{(n_{\max})}| = 3N(n_{\max})$, the parameterisation gives $3N(n_{\max}) \cdot 4 + 1 = 12N(n_{\max}) + 1$ symbolic indeterminates, naturality on the morphism panel reduces the system to nullity $3N(n_{\max})$ via the same closed-form argument (every η_{V_g} solves to unipotent upper triangular with one free parameter q_g; $\eta_T = 1$ falls out from the unit pair). The bit-exact `sympy`-rational verification at $n_{\max} \in \{1, 2, 3, 4\}$:

$n_{\max}$	$3N(n_{\max})$	morphisms	constraints	nullity	wall (s)
1	6	49	132	**6**	~ 1
2	15	271	736	**15**	1.85
3	27	811	2,296	**27**	12.23
4	42	1,891	5,461	**42**	55.74

Wall time grows polynomially in $n_{\max}$; the structural pattern is not isolated to the smallest non-trivial cutoff but $n_{\max}$-uniform. Φ-recovery is $3N(n_{\max})/3N(n_{\max})$ bit-exact at every cutoff. □ □

Remark 5.12 (Rank-nullity closed form)**.** The rank of the naturality + unit constraint matrix is $\mathrm{rank}(A) = 3 \cdot 3N(n_{\max}) + 1 = 9N(n_{\max}) + 1$ exactly (verified values: $19, 46, 82, 127$; equivalently forced by rank–nullity from the $3N$ nullity recorded in Theorem 5.7's proof, which gives $\mathrm{rank}(A) = (12N + 1) - 3N$). This matches the structural picture: naturality forces $p_g = 1$,

$r_g = 0$, $s_g = 1$ for each g (three linear constraints per generator) plus the unit normalisation $\eta_T = 1$, totalling $9N(n_{\max}) + 1$ *symbol fixings* on the $12N(n_{\max}) + 1$ indeterminates. All of them are independent over the morphism panel data, so the solution space has dimension $(12N+1) - (9N+1) = 3N$—which is the rank pattern above, by rank–nullity. The closed-form generalisation across all $n_{\max} \geq 1$ is conjectured and likely sprint-scale to verify symbolically; TC-2c does not pursue this sharpening.

5.8 Pro-system cofiltered coherence (TC-2d)

The per-cutoff equalities of Theorems 5.7, 5.9, and 5.11 package into a single pro-system statement via the PS-2 cofiltered functoriality of Section 2.3. Sprint Q5'-Tannakian-Closure TC-2d, 2026-06-06; memo `debug/sprint_q5p_tc2d_cofiltered_coherence_memo.md`, driver `debug/compute_q5p_tc2` tests `tests/test_tannakian_cofiltered_coherence.py`.

Theorem 5.13 (TC-2d, cofiltered coherence on $\mathrm{Aut}^{\otimes}(\omega)$)**.** *For every pair $k \leq m$, the Spec-dual of the PS-2 Hopf-algebra transition $\Phi_{m,k} : \mathcal{H}_{\mathrm{GV}}(m) \to \mathcal{H}_{\mathrm{GV}}(k)$ is a restriction map*

$$\rho_{m,k} : \mathrm{Aut}^{\otimes}(\omega)^{(m)} \to \mathrm{Aut}^{\otimes}(\omega)^{(k)},$$

which acts at the parameter level as coordinate truncation $t^{(m)} \mapsto t^{(m)}\big|_{P^{(k)}}$ on $\mathbb{Q}^{3N(m)}$. The restriction is a group homomorphism, satisfies the cofiltered transitivity $\rho_{n,k} = \rho_{m,k} \circ \rho_{n,m}$ at every triple $k \leq m \leq n$, and is bit-exact on every witness rep V_g for $g \in P^{(k)} \subseteq P^{(m)}$.

Proof. At the parameter level the restriction is the obvious dict truncation. On a witness rep V_g with $g \in P^{(k)}$, $\Phi(t^{(m)})(V_g) = \exp(t_g^{(m)} E_{12})$ (since $X_h^{V_g} = 0$ for $h \neq g$) is the same as $\Phi(t^{(m)}\big|_{P^{(k)}})(V_g)$ because the g-component of $t^{(m)}\big|_{P^{(k)}}$ equals the g-component of $t^{(m)}$. Group-hom property follows from $\Phi(t_1)\Phi(t_2) = \Phi(t_1 + t_2)$ (abelian unipotent factor) and $\mathbb{Q}$-linearity of parameter truncation. Transitivity at the parameter level reduces to $t|_{P^{(k)}} = (t|_{P^{(m)}})|_{P^{(k)}}$, a trivial set-theoretic identity for $P^{(k)} \subseteq P^{(m)} \subseteq P^{(n)}$.

The bit-exact verification: Block A (panel-level restriction coherence on V_g) tested at 6 pairs $(k, m) \in \{(1,2), (1,3), (2,3), (1,4), (2,4), (3,4)\}$ across 3 test parameter seeds each, totalling 225 residuals; Block B (transitivity at 4 triples $\times 3$ seeds) totalling 12; Block C (group-hom at 3 pairs $\times 2$ generators each) totalling 6. All 243 residuals are bit-exact zero. □ □

Corollary 5.14 (Inverse-limit Tannakian dual at the panel level)**.** *The cofiltered system $\{\mathrm{Aut}^{\otimes}(\omega)^{(n)}\}_{n \in \mathbb{N}}$ with restriction maps $\{\rho_{m,k}\}_{k \leq m}$ has inverse limit*

$$\mathrm{Aut}^{\otimes}(\omega)^{(\infty)} = \varprojlim_n \mathrm{Aut}^{\otimes}(\omega)^{(n)} = \varprojlim_n \big(\mathbb{G}_a^{3N(n)} \rtimes SL_2\big) = \mathbb{G}_a^{\infty} \rtimes SL_2,$$

at the natural-substrate panel level. The remaining content of the full long-range reconstruction is the pro-finite Tannakian theorem [2, 3] coherent with v3.66.0 FO3 Interpretation C at the period level on $\mathcal{O}_\infty$ (PS-3) of Section 2.4 — a structural identification between the unipotent part $\mathbb{G}_a^{\infty}$ of $\mathbb{G}_a^{\infty} \rtimes SL_2$ and the motivic Galois group on the mixed-Tate period ring $F(s)$ of $\mathcal{O}_\infty$ (SL_2, being perfect, maps trivially into a solvable group).

6 The level-4 cosmic-Galois comparison homomorphism

The per-cutoff equality of Corollary 5.14 identifies the Tannakian dual $\mathrm{Aut}^{\otimes}(\omega)^{(\infty)}$ with $\mathbb{G}_a^{\infty} \rtimes SL_2$ as a pro-algebraic group, but does not yet compare it to a published motivic-Galois target. This section gives that comparison: we exhibit an explicit homomorphism of pro-algebraic groups $\Phi^{\mathrm{inj}} : \mathcal{U}_4 \to \mathbb{G}_a^{3N}$ from the level-4 unipotent radical into the abelian factor of U_{GV}^*, factoring through $\mathcal{U}_4^{\mathrm{ab}}$ (Reading A; see Theorem 6.2), where $\mathcal{U}_4$ is the pro-unipotent radical of the level-4 motivic Galois group $\mathcal{G}_4 = \mathcal{G}_{\mathrm{MT}(\mathbb{Z}[i,1/2])}$ of Deligne 2010 [7] and Glanois 2015 [8]. The four pieces

of the construction — the period-level map, its multiplicativity, its coproduct compatibility, and the SL_2 factor's behaviour under the level-4 cyclotomic character χ_4 — are all in the literature; the assembly is the work of this section.

6.1 The period map

The level-4 motivic Galois group has Levi decomposition

$$\mathcal{G}_4 \;=\; \mathbb{G}_m \ltimes \mathcal{U}_4,$$

where $\mathbb{G}_m$ acts on each weight piece by the cyclotomic character χ_4 and $\mathcal{U}_4$ is the pro-unipotent radical generated by depth-graded motivic MZVs $\{\zeta^{\mathfrak{m}}(s_1,\dots,s_k)\}$ at level $N=4$ (Deligne 2010 [7]; Glanois 2015 explicit basis $\mathcal{B}^4$ [8]). The natural action of $\mathcal{G}_4$ on periods is faithful at level 4. This is not a general fact and we do not state it as one: faithfulness holds only in the exceptional cases $N \in \{2,3,4,6,8\}$ (Deligne 2010 [7], §4 *Fidelité*), together with $N=1$ (Brown 2012), and it *fails* for prime $N \geq 5$, where Goncharov showed the action is already unfaithful on $\wedge^2\mathfrak{u}^1_\omega$. Level 4 is one of the exceptional cases, which is all this paper needs; the underlying mixed-Tate package is Deligne–Goncharov [26]. Affine group schemes and commutative Hopf algebras are anti-equivalent, so a Hopf map out of $\mathcal{H}_{\mathrm{GV}}$ into the de Rham motivic Hopf algebra $\mathcal{A}^{(4)} \cong \mathcal{O}(\mathcal{U}_4)$ dualises to a homomorphism *from* $\mathcal{U}_4$ *into* $\operatorname{Spec}\mathcal{H}_{\mathrm{GV}}$; Theorem 6.2 states the map in that direction.

Concretely, the period map evaluates GeoVac's primitive Hopf generators on their structural period content (Paper 55).

Definition 6.1. The *level-4 period map*

$$\pi:\ \mathcal{H}_{\mathrm{GV}} \longrightarrow \mathrm{MT}(\mathbb{Z}[i,1/2])$$

sends a primitive generator $x_{(n,l),k}$ of $\mathcal{H}_{\mathrm{GV}}$ to its image under the master Mellin engine evaluator $\mathcal{M}[\mathrm{Tr}(D^k\,e^{-tD^2})]$ at the Camporesi–Higuchi spectrum on S^3 (Paper 32 §VIII; Paper 18 §III.7):

$$\pi(x_{(n,l),0}) \;=\; m_1^{(n,l)} \in M_1 \;\subset\; \mathbb{Q}[\pi,\pi^{-1}],$$

$$\pi(x_{(n,l),2}) \;=\; m_2^{(n,l)} \in M_2 \;\subset\; \bigoplus_{j\geq 0}\pi^{2j}\cdot\mathbb{Q},$$

$$\pi(x_{(n,l),1}) \;=\; m_3^{(n,l)} \in M_3 \;\subset\; \mathrm{MT}(\mathbb{Z}[i,1/2]),$$

where the $k=1$ (M3) evaluation carries the sector's *full* first-order period content — both the ungraded trace and the proven χ_{-4} vertex-parity grading $\mathrm{Tr}((-1)^N D\,e^{-tD^2})$ (Paper 28 [28] Theorem 3), so $m_3^{(n,l)}$ records which quarter-integer Hurwitz shift-class the sector feeds (3/4 if its Camporesi–Higuchi shell is even, 5/4 if odd; Remark 6.6). The map is extended multiplicatively to all of $\mathcal{H}_{\mathrm{GV}} = \mathrm{Sym}_{\mathbb{Q}}(V)$ (Paper 55 Theorems 3.2, 4.1, and 5.1 establish the three target inclusions).

The three target rings M_1, M_2, M_3 sit inside a common ambient cyclotomic mixed-Tate ring at level ≤ 4 over $\mathbb{Z}[i,1/2]$ (Paper 55 §6 role-disjointness lemma); this is the period content of $\mathcal{G}_4$ at the levels reachable by GeoVac's natural three-bullet partition.

6.2 The comparison homomorphism theorem

Theorem 6.2 (Level-4 depth-1 comparison)**.** *Let $\mathcal{A}^{(4)}$ be the de Rham motivic Hopf algebra at level 4 (Brown's $\mathcal{A}$ in the level-4 setting, $\mathcal{A}^{(4)} \cong \mathcal{O}(\mathcal{U}_4)$, in which $\zeta^{\mathfrak{a}}(2)=0$). Let $\rho:\ V \to \mathcal{A}^{(4)}$ send each M3 generator $x_{(n,l),1}$ to the depth-1 de Rham class of the motivic lift of its period (Definition 6.1; level-4 depth-1 lifts, Deligne 2010 [7]) and each M1 and M2 generator to 0. Then:*

1. ***(C1) Multiplicativity.*** *ρ extends uniquely to an algebra map $\mathcal{H}_{\mathrm{GV}} = \mathrm{Sym}_{\mathbb{Q}}(V) \to \mathcal{A}^{(4)}$. This is the universal property of a free commutative algebra and carries no further content.*

2. ***(C2) Coproduct compatibility.*** *The values of ρ on generators are primitive in $\mathcal{A}^{(4)}$, so the extension is a map of Hopf algebras.*

3. ***(C3) Dual homomorphism.*** *Dually it is a homomorphism of pro-algebraic groups*

$$\Phi^{\mathrm{inj}} : \mathcal{U}_4 \longrightarrow \operatorname{Spec} \mathcal{H}_{\mathrm{GV}} = \mathbb{G}_a^{3N},$$

   from *the unipotent radical of the level-4 motivic Galois group* into *the abelian factor of U_{GV}^*. Since the target is abelian it factors through $\mathcal{U}_4^{\mathrm{ab}}$. The SL_2 factor of U_{GV}^* does not enter.*

4. ***(C4) Rank.*** *The image of Φ^{inj} has dimension at most the rank of the M3 period Gram matrix: 2 per weight, the two χ_{-4} vertex-parity classes (Remark 6.6); the parity-blind restriction has rank 1. The M1 and M2 columns contribute nothing. The Gram rank is computed on the real periods, so it bounds the motivic rank from above and does not compute it.*

The full group $\mathcal{G}_4$ admits no such comparison: $\mathrm{Hom}(\mathcal{G}_4, \mathbb{G}_a) = 0$ *($\mathbb{G}_m$ acts on $\mathcal{U}_4^{\mathrm{ab}}$ with non-zero weights only, and* $\mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_a) = 0$*), so $\mathcal{O}(\mathcal{G}_4)$ has no non-zero primitives to receive V; and the M1 and M2 values are not primitive ($\Delta\pi^{2j} = \pi^{2j} \otimes 1$ in Brown's coaction), so they have rank 0 in $\mathcal{A}^{(4)}$.*

Proof. (C1) is the universal property of $\mathrm{Sym}_{\mathbb{Q}}(V)$. (C2): depth-1 classes of level-4 motivic periods (single polylogarithms at fourth roots of unity, equivalently the depth-1 part gr_D^1 dual to $\mathcal{U}_4^{\mathrm{ab}}$) are primitive in $\mathcal{A}^{(4)}$, because the de Rham logarithm of a root of unity vanishes; 0 is primitive. A primitively generated commutative algebra map that sends primitives to primitives is a Hopf map. (C3) is the anti-equivalence between commutative Hopf algebras and affine group schemes; a homomorphism into an abelian group kills the commutator subgroup. (C4): the image of Φ^{inj} is cut out by the linear span of $\rho(V)$ in the primitives of $\mathcal{A}^{(4)}$, whose dimension per weight is at most the number of $\mathbb{Q}$-linearly independent real M3 periods, which the Gram computation bounds by 2 (`tests/test_paper56_m3_parity_rank.py`; parity-blind rank 1, `tests/test_paper56_injection_g4_periodmap.py`). Pure-Tate values (π^{2j}, and the $\beta(\mathrm{odd}) \in \pi^{\mathrm{odd}}\mathbb{Q}$ part of an M3 value) vanish in $\mathcal{A}^{(4)}$. □ □

Remark 6.3 (Why level 4 and not level 1)**.** The refinement to $\mathcal{G}_4$ (rather than Brown's $\mathcal{G}_{\mathrm{MT}(\mathbb{Z})}$ at level 1) is motivated by Eskandari–Murty–Nemoto 2025 [25]: Catalan $G = \beta(2)$ *is* mixed Tate over $\mathbb{Q}(i) = \mathbb{Q}[i, 1/2]^{\times \otimes \mathbb{Q}}$ at level 4. *Honest scope:* the converse—that G is not a period of mixed Tate motives over $\mathbb{Q}$—is expected but, to our knowledge, open; the level-4 choice therefore rests on where G is *known* to live, not on a proven exclusion at level 1. Since GeoVac's M3 vertex-parity sector produces G in QED vertex sums (Paper 28 Theorem 3, Sub-sector 2), the natural cosmic-Galois target must contain at least level 4. No level ≥ 5 is needed: Paper 55 §5 shows that every M3 output on Camporesi–Higuchi S^3 is a Hurwitz value at quarter-integer shift $\{1/4, 3/4, 5/4\}$, which the explicit Glanois 2015 basis realises inside $\mathrm{MT}(\mathbb{Z}[i, 1/2])$.

Remark 6.4 (Reading-A abelianization at depth 1)**.** The comparison homomorphism of Theorem 6.2 sees only the *abelianization* of $\mathcal{U}_4$: Φ^{inj} factors through $\mathcal{U}_4^{\mathrm{ab}}$, the quotient dual to the depth-1 part gr_D^1, since GeoVac's generators are primitive and can only be matched with primitive (depth-1) motivic classes (proof of (C2) above). This matches the NA-1 structural finding (Section 8.3, Reading A): GeoVac's primitive Hopf substrate gives an abelian unipotent radical by the Cartier–Milnor–Moore theorem ([27] as recalled in Brown 2012 [3] §5), so the natural Tannakian dual $\mathbb{G}_a^{\infty}$ is the abelianization of any free non-abelian unipotent target. The depth-1 sub-quotient of $\mathcal{U}_4$ is exactly the abelianization $\mathcal{U}_4^{\mathrm{ab}} \cong \mathrm{gr}_D^1 \mathcal{U}_4$, and this is what Φ^{inj} sees on the unipotent side.

Remark 6.5 (Honest scope — partiality)**.** Three honest caveats refine the partial nature of the comparison homomorphism (Track C scoping memo, end-of-day 2026-06-06 synthesis):

- *M1 and M2 columns drop out.* Their values lie in $\mathbb{Q}[\pi, \pi^{-1}]$ and $\bigoplus_k \pi^{2k}\mathbb{Q}$, which are not primitive in the motivic coaction and vanish in $\mathcal{A}^{(4)}$ ($\zeta^{\mathfrak{a}}(2) = 0$). They therefore contribute nothing to the comparison (rank 0), which reflects GeoVac's three-bullet master Mellin engine partition (Paper 18 §III.7): M1 and M2 are pure-Tate and sector-blind.

- *The M3 column is rank* 2 *(parity-blind restriction rank-1).* The M3 column ($k = 1$, vertex-parity Hurwitz) is the only column mapping to a depth-graded target with non-trivial cyclotomic content. Its parity-blind per-sector content is the single scalar $\eta_{(n,l)}$ (Paper 55), under which each sector's period vector is $\eta_{(n,l)}$ times a shared Hurwitz block — collinear, Gram rank 1, det 0 (`tests/test_paper56_injection_g4_periodmap.py`). The faithful Deligne 2010 action on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_4)$ would give *full* injectivity only for a rank-N M3 period map. The *enriched* map (Definition 6.1) adds the proven χ_{-4} vertex parity — the per-(n, l) parity decomposition of $D(s)$ — which sorts sectors into the two shift-classes $3/4, 5/4$ and raises the M3 rank to 2 (Remark 6.6, `tests/test_paper56_m3_parity_rank.py`), capped there. So the image of Φ^{inj} in $\mathbb{G}_a^{3N}$ has dimension at most 2 per weight (Reading A).

- *Depth-blindness on the GeoVac side.* The comparison homomorphism is constructed at depth 1 (linear combinations of single Hurwitz values). GeoVac's primitive Hopf algebra cannot detect shuffle / deconcatenation structure (Reading A, Section 8.3), so the map cannot be promoted to a depth-respecting injection without enriching the substrate to the cofree-cocommutative shuffle Hopf algebra $T(V)$. Whether such an enrichment is structurally required (Reading B wins NA-1 sprint) or whether the abelianization is the correct geometric realisation (Reading A wins) is the genuinely open question pending the sprint-scale NA-1 depth-2 Mellin test.

Remark 6.6 (Vertex-parity resolution of the M3 column)**.** The M3 column's parity-*blind* restriction is rank-1 — its per-sector content is the single scalar $\eta_{(n,l)}$ (Remark 6.5). This remark backs the rank-2 statement of Theorem 6.2(C4). The follow-on — a "sector-resolved rank-≥ 2 M3 period map, a per-(n, l) parity decomposition of $D(s)$" — is supplied by the *proven* χ_{-4} vertex-parity identity (Paper 28 [28] Theorem 3): even Camporesi–Higuchi shells contribute at Hurwitz shift $3/4$ and odd shells at $5/4$, with $D_{\mathrm{even}}(s) - D_{\mathrm{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$ pure level-4 Dirichlet-L content. Each Coulomb sector (n, l) sits in its shell's parity class, so the parity-resolved period vector is $\eta_{(n,l)}$ times the basis vector of its shift-class, and the Gram rank equals the number of populated shift-classes: **rank 2 at every** $n_{\max} \geq 2$ (rank 1 at $n_{\max} = 1$, where only odd shells exist), bit-exact under both sector↔shell offset conventions (`tests/test_paper56_m3_parity_rank.py`). The two shift-classes are a genuine independent period, not a manufactured sign: $D_{\mathrm{even}}(4)$ and $D_{\mathrm{odd}}(4)$ are $\mathbb{Q}$-linearly independent in $\{\pi^2, \pi^4, G, \beta(4)\}$ as coefficient vectors (as numbers this rests on the conjectured independence of those constants; even the irrationality of G is open).

Two consequences. First, this is why Theorem 6.2(C4) records rank 2 (not the parity-blind rank-1): the M3 image is 2-dimensional — the two shift-classes $\{3/4, 5/4\}$, equivalently the two motivic *levels* (level-2 un-restricted $D(s) \oplus$ level-4 χ_{-4} difference) in the sum/difference basis. Second — sharpening the follow-on's implicit hope that "rank-$\geq 2 \Rightarrow$ injectivity" — the parity route is *capped* at rank 2 (there are only two parity classes), so it *cannot* deliver a faithful closed immersion (rank N) for $N > 2$: GeoVac's abelianized image is strict and *rank-2*, neither rank-1 nor faithful.

Scope. The χ_{-4}-graded trace $\mathrm{Tr}((-1)^N D\, e^{-tD^2})$ is admitted into the period map (Definition 6.1) as genuine first-order period content — it is the same $k = 1$ Mellin slot and a proven GeoVac observable (Paper 28 [28] Theorem 3). Under this enriched map the M3 column is rank

2, which is the rank recorded by Theorem 6.2(C4); the parity-blind restriction to the ungraded trace alone is the rank-1 sub-statement.

6.3 Numerical verification panel

The four compatibilities (C1)–(C4) admit explicit bit-exact verifications at every finite cutoff $n_{\max} \in \{1, 2, 3, 4\}$. The driver, data, memo, and regression tests are:

- Driver: `tests/test_paper56_injection_g4.py`.
- Memos: `debug/sprint_injection_g4_memo.md` (foundational sprint, $n_{\max} = 2$); `debug/sprint_injectio` (this section's extension to $n_{\max} \in \{3, 4\}$).

The (C1) and (C2) panels below hold by definition and test nothing about the motivic side, and the (C3) $\det g = 1$ panel bears on no part of the restated theorem; only the (C4) Gram rank does. The four panels have closed-form per-cutoff residual counts that depend only on $N(n_{\max}) = n_{\max}(n_{\max} + 3)/2$:

- *(C1) Multiplicativity panel.* For every ordered pair (x, y) of primitive generators of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$, the formal multiplicativity $\pi(xy) = \pi(x)\,\pi(y)$ holds in $\mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ by the universal property of the symmetric algebra. Closed form: $(3N(n_{\max}))^2$ bit-exact zero residuals.
- *(C2) Coproduct panel.* For every primitive generator x in $\mathcal{H}_{\mathrm{GV}}(n_{\max})$, the equality $\Delta x = x \otimes 1 + 1 \otimes x$ at depth 1 is bit-exact by definition of the primitive coproduct. Closed form: $3N(n_{\max})$ bit-exact zero residuals.
- *(C3) SL_2-to-Levi panel.* For every g in the five-element $SL_2(\mathbb{Q})$ panel of Theorem 5.2 (identity, unipotent, torus, generic, Weyl), $\det(g) = 1$ bit-exact. The SL_2-to-Levi check is $n_{\max}$-independent because the cyclotomic character χ_4 acts on the reductive factor and is blind to the $n_{\max}$-axis substrate; closed form 5 bit-exact zero residuals per cutoff.
- *(C4) M3 period-map panel (abelianized; injectivity refuted).* For each of the $N(n_{\max})$ M3-column generators of $\mathcal{H}_{\mathrm{GV}}(n_{\max})$ (the $k = 1$ Mellin slot), the period-side image is the single per-sector scalar $\eta_{(n,l)}$ (Paper 55) times the shared Hurwitz block in $\{\zeta(s, 1/4), \zeta(s, 3/4), \zeta(s, 5/4)\}$. Deligne 2010 faithfulness on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_4)$ would imply *full* $\mathbb{Q}$-linear independence of all N generators only for a rank-N per-sector map; the χ_{-4}-enriched content is rank 2 (parity-blind rank-1). Computing (rather than assuming) the $N(n_{\max}) \times N(n_{\max})$ Gram matrix of the parity-blind period-coefficient vectors gives **rank 1,** $\det = 0$ at every cutoff (`tests/test_paper56_injection_g4_periodmap.py`); the enriched χ_{-4} map is **rank** 2 (`tests/test_paper56_m3_parity_rank.py`; Remark 6.6), still $\ll N$, so the M3 column is not injective and lands in the abelianized image. (The earlier panel asserted a unimodular "standard basis" Gram with $\det = 1$; that was the hardcoded $\mathrm{eye}(N)$ the genuine computation refutes.) The depth-1 realisation of each generator in the Glanois basis $\mathcal{B}^4$ via the three quarter-integer Hurwitz blocks, with the Eskandari–Murty–Nemoto level-4 placement, remains valid as the description of the (collapsed) abelianized image.

Summing across the four panels gives the closed-form per-cutoff G4-Inj residual count

$$R^{\text{G4-Inj}}(n_{\max}) \;=\; (3N(n_{\max}))^2 + 3N(n_{\max}) + 5 + (1 + 3N(n_{\max})) \;=\; 9N(n_{\max})^2 + 6N(n_{\max}) + 6,$$

which evaluates to

$$R^{\text{G4-Inj}}(1) = 54, \quad R^{\text{G4-Inj}}(2) = 261, \quad R^{\text{G4-Inj}}(3) = 789, \quad R^{\text{G4-Inj}}(4) = 1854.$$

The per-cutoff breakdown is summarised in Table 2.

$n_{\max}$	$N(n_{\max})$	C1	C2	C3	C4	Total
1	2	36	6	5	7	54
2	5	225	15	5	16	261
3	9	729	27	5	28	789
4	14	1764	42	5	43	1854
	G4-Inj sum at $n_{\max} \in \{2,3,4\}$:					**2,904**

Table 2: Closed-form bit-exact residual counts for the level-4 cosmic-Galois comparison-homomorphism panel at every finite cutoff $n_{\max} \in \{1,2,3,4\}$. The sum over the production grid $\{2,3,4\}$ contributes 2,904 residuals to the master verification panel (Table 3); the C4 row counts the *abelianized* M3 panel (its closed-immersion injectivity is refuted — Theorem 6.2). The counts are the symbolic outputs of `tests/test_paper56_injection_g4.py`; the checks behind them hold by construction (C1–C3) or count the panel (C4), so the block is bookkeeping for the homomorphism, not an independent test of it. The closed-form $R^{\text{G4-Inj}}(n_{\max}) = 9N(n_{\max})^2 + 6N(n_{\max}) + 6$ matches each row bit-exact.

The extension from $n_{\max} = 2$ to $n_{\max} \in \{3,4\}$ is structural rather than merely numerical: the closed-form per-cutoff identity makes the panel size depend on $N(n_{\max})$ only, with no $n_{\max}$-specific structural surprises. The four compatibilities hold at every cutoff in the test grid, as they must by construction. The pattern is $n_{\max}$-uniform, mirroring the cofiltered coherence of TC-2d (Theorem 5.13).

6.4 The natural realisation is Hodge / Hodge-de-Rham

The comparison homomorphism $\Phi^{\text{inj}} : \mathcal{U}_4 \to \mathbb{G}_a^{3N}$ of Theorem 6.2 factors through $\mathcal{U}_4^{\text{ab}}$ and has image of dimension at most 2 per weight (the M3 column; M1 and M2 contribute nothing). The motivic Galois group $\mathcal{G}_4$ admits multiple realisations — Betti, algebraic de Rham, Hodge / Hodge-de-Rham, étale, ℓ-adic, crystalline — and the comparison map at the period-ring level is realisation-dependent.

We adopt as the standing convention that the natural target for GeoVac's comparison program is the *Hodge / Hodge-de-Rham realisation*, not the étale or ℓ-adic realisations. Two structural reasons make this the only consistent choice:

1. **Substrate compatibility.** The real structure J with $J^2 = -\mathbf{1}$ (the Kramers / $SU(2) = S^3$ quaternionic structure) exists bit-exact at finite cutoff and supplies the complex structure a real-Hodge / Hodge-de-Rham realisation requires. We flag a finite-cutoff caveat (Sprint Hodge-SL_2, `debug/sprint_hodge_sl2_memo.md`): the clean odd KO-dimension-3 relation $JD = +DJ$ is *not* bit-exact on the natural substrate. With CG-weighted adjacency the off-diagonal part of D anticommutes with J exactly ($\{J, A_{CG}\} = 0$), but the m_j-independent diagonal Camporesi–Higuchi part *commutes* with J, so $D = \Lambda + \kappa A$ satisfies neither $JD = +DJ$ nor $JD = -DJ$ (the residual is exactly $2\Lambda J$). The complex structure is thus present — the structural input the Hodge realisation requires — while the clean odd KO-dimension-3 real spectral triple is only partially realised at finite cutoff. No comparable prerequisite for an étale or ℓ-adic realisation has been verified, nor is there a natural candidate at finite cutoff.

2. **Absence of p-adic input.** GeoVac has no p-adic place data, no Frobenius action, and no Galois group of a number field acting on the substrate. Étale / ℓ-adic realisations of motivic Galois groups require precisely this kind of arithmetic input; GeoVac supplies none of it. Categorically, those realisations are unreachable from GeoVac's data.

The level-4 comparison homomorphism $\Phi^{\mathrm{inj}} : \mathcal{U}_4 \to \mathbb{G}_a^{3N}$ of Theorem 6.2 is therefore to be read as an *abelianized* de Rham homomorphism (Reading A; image of dimension at most 2 per weight). The Hain–Brown relative completion of $SL_2(\mathbb{Z})$ carries a canonical mixed Hodge structure (Hain 2014 [6], Theorem 3.8); any future upgrade of the comparison target from $\mathcal{G}_4$ to a sub-quotient of the Hain–Brown relative completion would inherit that canonical MHS. At the time of writing, the Hain–Brown identification at the period level was probed at Sym^2 depth 1 and depth 2 and returned NEGATIVE (see `debug/sprint_hb_pslq_test_memo.md` and `debug/sprint_na1_depth2_mellin_memo.md`); the shape-match at the categorical level (extension $1 \to U \to G \to SL_2 \to 1$ with U pro-unipotent and tensor category generated by $\mathrm{Sym}^k V$) persists, but the comparison target remains $\mathcal{G}_4$ rather than Hain–Brown. The Hodge realisation choice is robust to either resolution.

From convention to identification: GeoVac realises a CM point. The Hodge realisation can be made concrete on the fundamental rep. Restricting the Kramers J to a $j = 1/2$ doublet of the lowest Dirac shell gives the canonical complex structure $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on $V_{\mathrm{fund}} = \mathbb{Q}^2$; with the SL_2-invariant symplectic form $Q = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ the Riemann form $QJ = I$ is positive-definite, so $(V_{\mathrm{fund}}, J, Q)$ is a genuine polarised weight-1 Hodge structure.

Proposition 6.7 (GeoVac realises the CM point; CM field $\mathbb{Q}(i)$)**.** *The Hodge group (the special Mumford–Tate group,* $\mathrm{MT} \cap SL_2$*) of GeoVac's canonical polarised weight-*1 *Hodge structure* $(V_{\mathrm{fund}}, J, Q)$ *is the norm-*1 *torus of* $\mathbb{Q}(i)$ *— a* 1*-dimensional CM torus inside* SL_2 — not *the full* SL_2*. (The full Mumford–Tate group adds only the weight scalars,* $\mathrm{Res}_{\mathbb{Q}(i)/\mathbb{Q}}\mathbb{G}_m$*; the CM conclusion is unchanged.)*

Proof. J is rational with $J^2 = -I$, so $\mathbb{Q}[J] \cong \mathbb{Q}[t]/(t^2+1) = \mathbb{Q}(i)$ and the Hodge circle $\cos t\, I + \sin t\, J$ generates the norm-1 torus $T_J = \{aI + bJ : a^2 + b^2 = 1\}$, with $\det(aI + bJ) = a^2 + b^2 = N_{\mathbb{Q}(i)/\mathbb{Q}}(a + bi)$. T_J is the $\mathbb{Q}$-Zariski closure of the Hodge circle (a 1-dimensional $\mathbb{Q}$-torus in which the real circle is dense) and preserves the definite Riemann form $QJ = I$, hence $T_J = SO(2)$ as a $\mathbb{Q}$-group, a proper 1-dimensional subgroup of SL_2 ($\dim T_J = 1 < 3 = \dim SL_2$). Equality of the Hodge group with SL_2 would require a non-algebraic (transcendental) complex structure, which the algebraic GeoVac substrate cannot produce. Verified bit-exact in `tests/test_paper56_hodge_sl2.py`. □

Remark 6.8 (Why this explains the Hain–Brown negative)**.** Proposition 6.7 upgrades the Hodge-realisation *convention* above to a concrete *identification*: GeoVac realises the complex-multiplication point of the weight-1 story, with CM field $\mathbb{Q}(i)$. This structurally explains the Hain–Brown period negative. Hain–Brown is the generic (full-SL_2, non-CM) modular locus, whose periods involve E_4, E_6, Δ; at a CM point the modular forms degenerate and the periods become CM-pure over the CM field. GeoVac's periods are therefore $\mathbb{Q}(i)$-pure (Catalan G, β-values), not modular — exactly the negative observed at Sym^2 depth 1 and depth 2.

Remark 6.9 (On S^3 the $\mathbb{Q}(i)$ triangulation is spin-forced)**.** The CM field $\mathbb{Q}(i)$ coincides with the level-4 field $\mathbb{Q}(\mu_4)$ of $\mathcal{G}_4 = \mathcal{G}_{MT(\mathbb{Z}[i,1/2])}$ and with the field of GeoVac's period content (Catalan G mixed-Tate over $\mathbb{Q}(i)$, Eskandari–Murty–Nemoto 2025 [25]). A structural audit (`debug/sprint_hodge_sl2_audit_qi_triangulation.py`) shows this is not a coincidence on S^3: both the Hodge CM field and the level-4 period content are gated there by the half-integer spin structure (the conductor-4 periods also arise from the metaplectic shift on S^5 without spin, Remark 6.10). The complex structure exists iff $\varepsilon = (-1)^{2j} = -1$ (spinor; scalar $(-1)^{2l} = +1$ gives no complex structure), and the β/Catalan period content carries conductor $4 = |\mathrm{disc}\,\mathbb{Q}(i)|$ via the Dirac vertex-parity character χ_{-4} (Paper 28 [28]), whereas the scalar S^3 Laplacian ζ is pure-Tate over $\mathbb{Q}$ (conductor 1). The scalar null removes $\mathbb{Q}(i)$ from both sides at once. The motivic identification level-4 $= \mathbb{Q}(\mu_4)$ is taken from Deligne 2010 [7]; the audit verifies the common spin cause and the same-field identity, not the motivic result.

Remark 6.10 (The posited wedge flow realises the Hodge circle; μ_4 at the quarter-periods)**.** The Hodge circle of Proposition 6.7 is not only abstractly present in the substrate — it is the compact flow of the posited wedge generator. On the $j = \frac{1}{2}$ doublet the wedge generator of the truncated modular analysis (Paper 42) is the posited $K = \mathrm{diag}(2m_j) = 2J_z$, with integer spectrum, so $e^{2\pi i K} = \mathbf{1}$ exactly — an automatic 2π closure; its identification with a Bisognano–Wichmann modular flow of a wedge-restricted state, and hence with a KMS $\beta = 2\pi$ circle, is open (Paper 42). Transported to the Kramers frame U (the eigenframe of J), the flow *is* the Hodge circle, $U\, e^{itK}\, U^\dagger = \cos t\, \mathbf{1} + \sin t\, J$; in particular J is the flow's $\beta/4$ quarter-period point, the four quarter-periods are $\{\mathbf{1}, J, -\mathbf{1}, -J\} \cong \mu_4$ — the conductor $4 = |\mathrm{disc}\, \mathbb{Q}(i)|$ is the order of the flow's torsion point — and the half-period separates the sectors: $e^{i\pi K} = -\mathbf{1}$ on the spinor doublet (half-integer m_j; the spin double cover, visible inside the compact circle) versus $+\mathbf{1}$ on any integer-l scalar sector, which factors through μ_2 and stays pure-Tate (Remark 6.9). All statements are exact (`tests/test_paper56_kms_hodge_circle.py`). Three qualifications keep this an Observation. First, any two maximal tori of $SU(2)$ are conjugate, and both halves of the statement are separately classical (the Hodge circle of a CM elliptic curve generates the norm-1 torus; the continuum Bisognano–Wichmann flow is KMS at $\beta = 2\pi$, prior art); the content is the identification of the program's two *specific* operators — one of them posited, not derived from a state — and the location of the arithmetic data (μ_4 torsion, spinor sign) at canonical points of the flow. Second, compactness alone does not produce $\mathbb{Q}(i)$: the scalar sector also generates a compact circle yet its periods are pure-Tate over $\mathbb{Q}$. The discriminating datum is, however, *not* merely whether the μ_4 point acts faithfully (spinor) or through μ_2 (scalar): the Bargmann–Segal Hardy phase circle of the harmonic oscillator (Paper 24) carries a *faithful* μ_4—the metaplectic zero-point $\frac{3}{2}$ lifts the substrate's bipartition parity to order four—yet admits no CM Hodge structure, because its generator $N + \frac{3}{2}$ is strictly positive (no $\pm$-paired spectrum, hence no traceless 2-plane for a $\mathbb{Q}$-rational J to rotate). The discriminant separating the two carriers examined is a faithful μ_4 acting as a real rotation of a $\mathbb{Q}$-rational 2-plane, i.e. a $\pm$-symmetric generator spectrum—which the quaternionic spinor doublet supplies (Paper 24 reaches the same negative by three independent exact obstructions, of which this is one facet; no completeness or uniqueness claim is made). The conductor-4 *period* content, by contrast, does transfer: the Hardy tower's parity-graded spectral zeta is $2^{s-3}(\beta(s) - \beta(s-2))$, exactly one quarter of the Dirac value, with the χ_{-4} produced by the half-integer index shift times the parity grading (spin on S^3, metaplectic on S^5)—so an order-2 datum lifted to order 4 suffices for conductor-4 periods, while the CM *Hodge structure* additionally needs the $\pm$-paired rational carrier (`tests/test_paper24_hardy_circle_arithmetic.py`). Third, the 2π-closure re-reads an already-registered structural fact (the compactness of the finite-cutoff flow, Paper 45; the finite generator is not the continuum boost); it is a consistency statement, not independent evidence. We have not found this flow-level identification in the literature: the closest structures are the labelling of extremal KMS_∞ states by complex-multiplication points in the quantum-statistical-mechanics program of Connes–Marcolli–Ramachandran [30] (a labelling of states, not an identification of the flow) and a recent physical $U(1) \times U(1)$ R-symmetry flow realising a Hodge decomposition with complex multiplication at special loci [31] (an R-symmetry, not a thermal/modular flow).

Remark 6.11 (An independent chemistry realisation of the $\mathbb{Q}(i)$ arithmetic (Paper 59): convergent, not common-cause)**.** A concrete elliptic period from a wholly different corner of the program carries the same conductor-4 arithmetic — by a different forcing. The three-center electron-repulsion integral of Paper 59 [32] reduces, in momentum space, to a two-scale Bessel moment whose curve is the Legendre family over $X(2) = \Gamma(2)\backslash\mathfrak{H}^*$ (modulus $\lambda = 1 - \rho$, with ρ the ratio of the two densities' Fock scales); its modular and $\mathbb{Q}(i)$-CM structure below is developed in the companion Paper 61 [33]. The load-bearing arithmetic statement is fibre-independent: the fibre period is $K(\lambda) = \frac{\pi}{2}\,\theta_3^2$, and θ_3^2 *is* the theta series of $\mathbb{Z}[i]$ — by Jacobi's two-squares theorem $r_2(n) = 4\sum_{d|n} \chi_{-4}(d)$, the weight-one, level-4 Eisenstein series of charac-

ter χ_{-4}, with $L(\theta_3^2, s) = 4\zeta(s)\beta(s)$ — and the physical Bessel twist promotes the level from $\Gamma(2)$ to $\Gamma(4)$ (its decay rate makes $\sqrt{\rho} = \theta_4^2/\theta_3^2$ physical), whose field is $\mathbb{Q}(\mu_4) = \mathbb{Q}(i)$ (verified in `tests/test_paper59_theta_chi4.py`). The $\tau = i$ fibre at $\rho = \frac{1}{2}$, where the period is the $\mathbb{Q}(i)$-pure Γ-value $K(\frac{1}{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$ (Chowla–Selberg, verified $\sim 10^{-51}$), remains a clean instance but is not load-bearing: any family sweeping its full real modulus locus passes through infinitely many CM fibres (not every one — those with non-real λ, such as discriminant -3, are off the contour), the physically distinguished fibre is the coincident-scale cusp $\rho = 1$ rather than $\rho = \frac{1}{2}$, and landing on the Legendre family is itself expected for a four-branch-point curve (Paper 59's own deflation). The group side sharpens this: the only involution the physics induces on the modulus is $\rho \mapsto 1/\rho$ (the density exchange; the observable descends to $X_0(2)$, Paper 59), whose physical fixed locus is the pure-Tate cusp $\rho = 1$—so $\rho = \frac{1}{2}$ is one member of the exchange orbit $\{\frac{1}{2}, 2\}$, a visited fibre, not a fixed one. The two routes to $\mathbb{Q}(i)$ are therefore *convergent, not common-cause*: the Dirac route is spin-forced (half-integer shift × parity split → quarter-integer Hurwitz arguments → χ_{-4}, Remark 6.9) and mixed-Tate over $\mathbb{Q}(i)$, while the chemistry route is level-forced (the $\mathbb{Z}[i]$ theta series; $\Gamma(2) \to \Gamma(4)$) and mixed-elliptic with an irregular twist. They share the arithmetic pattern $4 = 2 \times 2$ — a double-cover datum combined with a second $\mathbb{Z}_2$ — but the two double covers belong to different groups, and no comparison map between the two underlying $\mathbb{Q}^2$-carriers has been exhibited (the weight-$\frac{1}{2}$/spin analogy has a classical anchor: theta characteristics *are* spin structures, Atiyah 1971 [29]). The convergence is now quantified both ways: across the corpus's period-identification record, *blind* searches land on disc -4 in 0 of 30 independent campaigns (including two that targeted it), while *forced* constructions that exit the pure-Tate ring at all land on it in 3 of 3 — $\mathbb{Q}(i)$ is the first CM field the substrate's rational arithmetic can reach, a constrained-landscape statement rather than a coincidence (CHANGELOG v4.103.0); and on the modular side the arithmetic content of any single $\mathbb{Z}_2$ realisation is provably one bit ($\mathrm{PSL}_2(\mathbb{Z}) \cong \mathbb{Z}_2 * \mathbb{Z}_3$ torsion: order two forces disc -4, order three disc -3). Two concrete upgrades would convert convergence into mechanism: a comparison map *natural in the GeoVac data* — e.g. presenting the fundamental doublet as H_1 of a GeoVac-constructed abelian variety; note the polarisation-matching clause is void as a discriminator, since $h(\mathbb{Q}(i)) = 1$ makes the principally polarised rank-2 $\mathbb{Q}(i)$-CM Hodge structure unique up to isomorphism, so even a tautological identification carries $QJ = I$ to the Riemann form and only *functoriality* can discriminate — or a measured $\beta(2)$ in the physical integrated observable itself. The second test has now been *run*: an exact refactorization of the observable (Paper 59, the (k, w) frame) certified its value to 66 cross-validated digits, and the guarded, decoy-calibrated search is **decisively negative** for $\beta(2)$/Catalan content at height ≤ 10 through weight three (64 working digits) — no measured conductor-4 content exists in the integrated observable at clean heights, so, at reachable heights, the conductor-4 arithmetic remains a demonstrated property of the *family* (θ_3^2) and of the $D \to 0$ shadow, not of the physical value. (Calibration rider, correcting the earlier "decidable at ≥ 32 digits": height budgets scale as $10^{D/n}$ in the ring dimension n, so the dimension-twenty ring needed ~ 64 digits for even a height-10 verdict; the negative excludes clean closed forms, not large-height ones.) With T-1 tautology-blocked and T-2 negative at its reachable height, the seam stands as *convergent* on all tested legs. Whether the integrated observable, or its non-CM fibres, reaches the generic Hain–Brown locus remains open.

6.5 The equality direction is long-range

Theorem 6.2 gives the depth-1 comparison $\mathcal{U}_4 \to \mathbb{G}_a^{3N}$. Its image is at most 2-dimensional per weight, while $\mathcal{U}_4^{\mathrm{ab}}$ has many more depth-1 generators per weight, so the map has a large kernel; a primitive Hopf substrate cannot see depth ≥ 2 at all. At depth ≥ 2 no GeoVac computation has yet produced an irreducible level-4 motivic MZV that is not reducible to depth-1 products. Whether an enriched GeoVac Hopf structure would receive such content is the long-range forward research question — testable in finite sprint-scale steps by exhaustion at increasing loop

order in the master Mellin engine.

7 Verification panel

Sub-track	Statement	Residuals
PS-1	Transitions $\Phi_{m,k}$ + cofiltered axiom (PS-1)	130
PS-2	U^*-equivariance ($\mathbb{G}_a$ + class-level) (PS-2)	485
PS-3	Inverse limit + continuity (PS-3)	284
PS-4	Endomorphism rigidity (PS-4)	872
TC-1a	$\mathrm{Rep}_{\mathrm{fin}}$ abelian	106
TC-1b	$\mathrm{Rep}_{\mathrm{fin}}$ symmetric monoidal	56
TC-1c	$\mathrm{Rep}_{\mathrm{fin}}$ rigid	50
TC-1d	Fiber functor ω exact + faithful + $\otimes$ + unit	40
TC-1e	$\mathbb{G}_a$ inclusion (natural $\otimes$-auto axioms + group law)	98
TC-1f	SL_2 axioms on $\mathfrak{PW}$ panel	135
TC-1f	$\mathbb{G}_a \times \mathrm{SL}_2$ commute on combined reps	100
TC-1f	Full-panel injectivity at $n_{\max} = 2$	255
TC-2a	Finite-cutoff equality on abelian factor at $n_{\max} = 2$	16
TC-2b	SL_2-factor equality (Sym^2 + det + decomp + dim)	17
TC-2c	Per-cutoff equality at $n_{\max} = 3, 4$ (Φ-recovery + dim + η_T)	73
TC-2d	Cofiltered coherence (panel + transitivity + group-hom)	243
G4-Inj	Comparison panels C1–C4, by construction ($n_{\max} = 2$)	261
G4-Inj	Same panel at $n_{\max} = 3$	789
G4-Inj	Same panel at $n_{\max} = 4$	1854
Total bit-exact zero residuals		**5,864**

Table 3: The full Tannakian-substrate verification panel: 5,864 bit-exact zero residuals across the four pro-system sub-tracks (PS-1/2/3/4), the six Tannakian-foundation sub-tracks (TC-1a/b/c/d/e/f), the four converse-direction sub-tracks (TC-2a/b/c/d), and the level-4 cosmic-Galois comparison-homomorphism panel extended to every finite cutoff in the production grid $n_{\max} \in \{2, 3, 4\}$ (G4-Inj $\times 3$). All residuals computed in `sympy`-rational arithmetic; no floating-point steps anywhere in the verification chain.

The full driver / data / memo trail of the verification panel is:

- PS-1: `debug/compute_q5p_ps1_transitions.py`, `debug/data/sprint_q5p_ps1_transitions.json`, `debug/sprint_q5p_ps1_transitions_memo.md`.
- PS-2: `debug/compute_q5p_ps2_ustar_compatibility.py`, `debug/data/sprint_q5p_ps2_ustar_compat` `debug/sprint_q5p_ps2_ustar_compatibility_memo.md`.
- PS-3: `debug/compute_q5p_ps3_inverse_limit.py`, `debug/data/sprint_q5p_ps3_inverse_limit.json` `debug/sprint_q5p_ps3_inverse_limit_memo.md`.
- PS-4: `debug/compute_q5p_ps4_endo_rigidity.py`, `debug/data/sprint_q5p_ps4_endo_rigidity.json` `debug/sprint_q5p_ps4_endo_rigidity_memo.md`.
- TC-1a/b/c/d/e/f: `debug/compute_q5p_tc1{a, b, c, d, e, f}_*.py`, `debug/data/sprint_q5p_tc1{a, b, c, d, e}_*.json` + `debug/data/q5p_tc1f_sl2_inclusion.json`, + memos in `debug/sprint_q5p_tc{b, c, d, e, f}_*.md`.

- TC-2a: `debug/compute_q5p_tc2a_aut_equality.py`, `debug/data/sprint_q5p_tc2a_aut_equality.jsc` `debug/sprint_q5p_tc2a_aut_equality_memo.md`.
- TC-2b: `debug/compute_q5p_tc2b_sl2_aut_equality.py`, `debug/data/sprint_q5p_tc2b_sl2_aut_equa` `debug/sprint_q5p_tc2b_sl2_aut_equality_memo.md`.
- TC-2c: `debug/compute_q5p_tc2c_higher_cutoff.py`, `debug/data/sprint_q5p_tc2c_higher_cutoff.j` `debug/sprint_q5p_tc2c_higher_cutoff_memo.md`.
- TC-2d: `debug/compute_q5p_tc2d_cofiltered_coherence.py`, `debug/data/sprint_q5p_tc2d_cofilter` `debug/sprint_q5p_tc2d_cofiltered_coherence_memo.md`.

The module `geovac/tannakian.py` (TC-1a..TC-1f, ∼2,400 lines) and `geovac/pro_system.py` (PS-1..PS-4) implement the substrate; both are bit-exact `sympy.Rational` / `Integer` throughout. Test coverage: 282 tests pass (49 base Tannakian + 30 rigidity + 24 fiber functor + 25 TC-1e + 60 TC-1f + 61 pro-system + 9 TC-2a + 14 TC-2b + 3 TC-2c fast + 7 TC-2d; plus 1 TC-2c slow, $n_{\max} = 4$, run separately).

8 Open questions and frontier

8.1 The converse direction — closed at finite cutoff (TC-2)

The converse direction of Deligne–Milne 1982 Theorem 2.11 on the finite-cutoff substrate, $\mathrm{Aut}^{\otimes}(\omega) = U^*_{\mathrm{Levi}}$ at each $n_{\max}$, is closed by the TC-2 sub-arc (Theorems 5.7, 5.9, 5.11, 5.13). The substantive open content is the *inverse-limit identification at the natural cosmic-Galois comparison target.*

8.2 The natural cosmic-Galois comparison target

The motivic group to compare with $\mathrm{Aut}^{\otimes}(\omega)^{(\infty)}$ is the level-4 motivic Galois group

$$\mathcal{G}_4 = \mathcal{G}_{MT(\mathbb{Z}[i,1/2])} = \mathbb{G}_m \ltimes \mathcal{U}_4,$$

of Deligne 2010 [7] and Glanois 2015 [8], *not* Brown's smaller $\mathcal{G}_{MT(\mathbb{Z})}$. Three published results motivate this refinement: Paper 55 [23] places its census of GeoVac's spectral periods inside $\mathrm{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4; Eskandari–Murty–Nemoto 2025 [25] construct a two-dimensional motive carrying Catalan $G = \beta(2)$, prove that *this motive* is not mixed Tate over $\mathbb{Q}$, and show it becomes mixed Tate after base change to $\mathbb{Q}(i)$—which is where G is known to live, and matters here because GeoVac's M3 vertex-parity sector produces G in QED vertex sums. (The stronger statement, that G is not a period of *any* mixed Tate motive over $\mathbb{Q}$, is expected but open, so this motivates rather than forces the level.) And Deligne 2010's faithfulness of the level-4 action on $\pi_1^{\mathrm{mot}}(\mathbb{G}_m - \mu_4)$ supplies the faithful action underlying the level-4 comparison. (Injectivity of that comparison would require a rank-N M3 period map, which Theorem 6.2 refutes; the χ_{-4} vertex parity caps the rank at 2.)

> **Theorem (depth-1 comparison, this paper).** Sending M3 generators to the depth-1 de Rham classes of their level-4 motivic periods and M1, M2 generators to 0 gives a Hopf map $\mathcal{H}_{\mathrm{GV}} \to \mathcal{A}^{(4)}$, dually a homomorphism $\mathcal{U}_4 \to \mathbb{G}_a^{3N}$ that factors through $\mathcal{U}_4^{\mathrm{ab}}$, with image of dimension at most 2 per weight (Theorem 6.2).
>
> **Open question (depth ≥ 2).** Whether an enriched GeoVac Hopf structure (non-primitive, e.g. the shuffle algebra $T(V)$) receives level-4 motivic content of depth ≥ 2.

The depth-1 comparison is a theorem in the direction $\mathcal{U}_4 \to \mathbb{G}_a^{3N}$ (Section 6). The open direction is depth ≥ 2: no depth-≥ 2 level-4 cyclotomic MZV has yet been observed in any GeoVac computation, and a primitive Hopf substrate could not receive one (Remark 6.5 delimits scope).

A common framing slip worth flagging: the two groups are related by a *surjection* $U_{CM}^* \twoheadrightarrow \mathcal{G}_{MT(\mathbb{Z})}$, not an identification — an elementary comparison of two free pro-unipotent groups rather than an imported theorem, and one Brown himself declines to assert. Connes–Marcolli's U_{CM}^* has one generator in every positive integer degree (free non-abelian) while $\mathcal{G}_{MT(\mathbb{Z})}$ has generators only in odd degrees ≥ 3; the surplus is the cancellation of even-zeta values $\zeta(2k) \in \pi^{2k}\mathbb{Q}$ already accounted for by the $\mathbb{G}_m$ factor. The GeoVac–cosmic-Galois comparison should be stated against $\mathcal{G}_4$, with the relationship to U_{CM}^* obtained via the (level-1) sub-quotient.

8.3 Abelian vs free non-abelian unipotent (NA-1 structural finding)

A structural feature distinguishes U_{GV}^* from the motivic-Galois target $\mathcal{U}_4$: GeoVac's unipotent radical is *abelian* ($\mathbb{G}_a^\infty$) while the motivic unipotent radical is *free non-abelian*. This is not an accidental commutation of generators; by the Cartier–Milnor–Moore theorem in characteristic 0, any primitive Hopf algebra $\mathrm{Sym}_{\mathbb{Q}}(V)$ has abelian Tannakian dual by construction. For non-abelian content the substrate would need to be enriched to the cofree-cocommutative shuffle Hopf algebra $T(V)$ with deconcatenation coproduct — the motivic-style structure (Brown 2012 [3]). Two structural readings remain consistent with current evidence:

- **Reading A (abelianization).** GeoVac respects the primitive coproduct, so the comparison sees only the *abelianization* of the motivic free non-abelian unipotent — a specific weight-only geometric realisation that drops the depth/iteration content.
- **Reading B (shuffle enrichment).** GeoVac's physical content (Mellin moments, master Mellin engine products) naturally respects the shuffle coproduct, and the substrate should be enriched to $T(V)$, at which point U_{GV}^* upgrades to a free non-abelian unipotent.

A sprint-scale concrete test distinguishes the readings: compute the joint Mellin transform of two specific depth-1 moments (M2 Seeley–DeWitt × M3 vertex-parity Hurwitz) and check whether the result factorises symmetrically (Reading A) or asymmetrically / deconcatenation-pair (Reading B). Sprint memo `debug/sprint_q5p_na1_non_abelian_probe_memo.md`.

Reading A in the even flavour only (Sprint JLO-Depth2, 2026-06-07). The naive Mellin-trace test above turns out to be *structurally insensitive* to the A/B distinction. Sprint NA-1 (2026-06-06) demonstrated that the depth-2 joint Mellin transform $J(s_1, s_2)$ collapses bit-exactly to a depth-1 parity-graded Mellin moment at shifted argument, $J(s_1, s_2) = M_3^{\gamma P}(s_1+s_2-1)$, on the natural Camporesi–Higuchi diagonal substrate (Paper 55 [23] Theorem `thm:na1_diagonal_collapse`). Sprint NA-1-offdiag (2026-06-06) generalised this: the trace-functional collapse holds on *any* substrate, commuting or non-commuting, because the trace functional is blind to off-diagonal entries of inserted operators in the D-eigenbasis (Paper 55 [23] Theorem `thm:na1_trace_functional_collapse`, "Reading C-strong"). Substrate enrichment cannot recover the A/B distinction at the trace-of-single-γ-insertion level.

The right next probe is the JLO entire cyclic cocycle at depth 2, $\chi_D(a_0, a_1, a_2)$, where two non-D-function insertions between three heat-kernel factors preserve the off-diagonal information the trace functional discards. Sprint JLO-Depth2 (2026-06-07) returns swap symmetry in both flavours and Reading A in the even flavour: on the truncated Camporesi–Higuchi spectral triple at $n_{\max} \in \{2, 3\}$ with the sector-idempotent algebra $\mathcal{A} = \mathbb{C}\langle e_0, \ldots, e_{N-1}\rangle$, the cocycle $\chi_D(a_0, a_1, a_2; t)$ is bit-exactly swap-symmetric at both leading order (t^0) and next order (t^1), in both the even (chirality-graded) and odd (plain trace) flavours. The 25-pair $\{1, e_s, e_t\}$

panel returns 40 bit-exact zero swap-asymmetric residuals at $n_{\max} = 2$; in the even flavour the full S_3-orbit test on $\chi_D(e_0, e_1, e_2)$ returns the same value across all six permutations, while in the odd flavour the orbit takes two values of opposite sign. See Paper 55 [23] Theorem `thm:jlo_depth2_reading_A`.

The even-flavour depth-2 content sits in the symmetric sector of $S_3 \curvearrowright \mathcal{A}^{\otimes 3}$; the odd flavour carries antisymmetric content. The honest-scope caveat: the swap symmetry is a consequence of D being real symmetric on this substrate (a complex Hermitian perturbation breaks it), not independent evidence beyond commutativity of $\mathcal{A}$, trace cyclicity and reality.

Strategic implication. On the natural substrate, GeoVac's JLO-cocycle-level content sees only the abelianisation-by-Cartier–Milnor–Moore of the cosmic-Galois universe. The map of Theorem 6.2, $\mathcal{U}_4 \to \mathbb{G}_a^{3N}$ through $\mathcal{U}_4^{\mathrm{ab}}$, is the statement of what is available, with image of dimension at most 2 per weight (the M3 column). Reading B (shuffle Hopf $T(V)$ enrichment) is *not* ruled out at depth 2: the even flavour is symmetric, but the odd flavour's S_3 orbit changes sign, so the probe does not decide between the readings. Two open questions remain: (i) non-commutative inner-factor enrichment of $\mathcal{A}$ could expose Reading B content at higher Mellin slot indices, but Sprint H1's Yukawa-non-selection theorem (Paper 32 [16] §VIII.C) already classifies the Yukawa *point* as calibration data (the moduli space and its dimension are forced; Paper 57), not autonomously generated by the GeoVac outer triple, so this is not a sprint-scale follow-on; (ii) whether depth-≥ 2 level-4 content reaches an enriched GeoVac Hopf structure remains the named long-range forward research question (§6.5), testable by exhausting GeoVac's M3 sector for level-4 cyclotomic motivic MZVs at depth ≥ 2.

8.4 Pro-finite assembly (long-range)

The full long-range content of the cosmic-Galois closure is the pro-finite Tannakian theorem coherent with v3.66.0 FO3 Interpretation C at the period level on $\mathcal{O}_\infty$ (PS-3, Theorem 2.5). It would relate the inverse-limit $\mathrm{Aut}^{\otimes}(\omega_{\mathcal{O}_\infty}) = \mathbb{G}_a^{\infty} \rtimes SL_2$ to $\mathcal{G}_4$ at the period level through a homomorphism from $\mathcal{U}_4$ into its unipotent part $\mathbb{G}_a^{\infty}$ (Theorem 6.2), in the spirit of Brown 2012 [3], Deligne 2010 [7], and Glanois 2015 [8].

8.5 Sprint-scale follow-ons

- **Higher-$n_{\max}$ injectivity panels.** Theorem 5.4 at $n_{\max} \in \{3, 4, 5\}$: same construction, panel size grows linearly in $n_{\max}$ as $3 \cdot N(n_{\max})$. Sprint-scale.
- **SL_2 on higher Peter–Weyl reps.** Theorem 5.2 on $\mathrm{Sym}^k V_{\mathrm{fund}}$ for $k \geq 3$. Sprint-scale.
- **Non-natural-substrate enlargements.** The Peter–Weyl decoration of §5.2 is the natural substrate (acts on disjoint tensor factor from $\mathbb{G}_a^{3N}$, so commutativity is by the Kronecker identity). Genuinely-new strong-form content lives on *enlarged* substrates where $\mathbb{G}_a$ and SL_2 act on the same tensor factor; that is long-range structural content, analogous to the Paper 46 Appendix B enlarged-substrate program.
- **Inverse-limit reconstruction.** The reconstruction argument at the inverse-limit level $\mathcal{O}_\infty$ coherent with the v3.66.0 FO3 period content is named as PS-3.5 in the roadmap; long-range, the heart of Tannakian reconstruction for GeoVac.

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